Additional file 1: Supplementary Material

Evaluation area and the limits of wildfire model transfer

Supplementary material for *Evaluation area and the limits of wildfire model transfer*.

Section S1 gives the sensitivity analyses and the details behind the Results. Section S2 gives the supporting tables, Section S3 the protocol details and limitations, Section S4 the target-label recovery curve, Section S5 the code and Section S6 the data. Tables are numbered S1 to S36, independently of the sections. Supplementary equations are numbered (S1), (S2) and so on. Every value is read from a released output, named by its path in the repository https://github.com/dryuemco/thermal-twin. Paths that begin with `paper/labelfix_rerun/` hold the corrected-label outputs. Values computed under the original Manavgat label are marked "original label".

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S1 Sensitivity analyses and detail

Each analysis below changes one design choice and keeps everything else fixed. None changes a conclusion of the main text. Section S1.6 is different: it tests a claim of the introduction, and the claim is not supported.

S1.1 Evia AOI and prevalence

The raw transfer analyses between Evia and Bejís, Manavgat and Muğla were repeated with the earlier, high-prevalence Evia rectangle. No qualitative conclusion changed. Thermal raw transfer AUCs moved by up to 0.134. The largest change was Evia to Manavgat, at 0.543 with the earlier rectangle against 0.677 with the extended one. No direction changed side of the chance line (`paper/labelfix_rerun/round7/r7f_evia_legacy.csv`). On the earlier rectangle the natural-vegetation population is smaller, and its burned prevalence is higher than on the extended rectangle (0.287; Table S15).

S1.2 CORAL regularisation

Nine λ values from 0 to 10^{-1} were tested on four directions (Bejís and Muğla, and Manavgat and Muğla, both ways). CORAL transfer AUC moved by at most 0.014 within any direction and model, and by at most 0.009 for the thermal model (`paper/labelfix_rerun/exports/coral_lambda_sensitivity_b74d643e/metrics.csv`). λ was not selected on performance.

The Manavgat and Bejís pair is not in this sweep. Under the original label, the adapted result of this pair depended on λ. It was therefore tested separately at $\lambda = 10^{-5}$, 10^{-3}, 10^{-1} and 1 with the authors' own re-implementation of the pipeline's CORAL (`paper/labelfix_rerun/step10/coral_lambda_sensitivity.csv`). This implementation differs slightly from the pipeline's: at $\lambda = 10^{-5}$ it gives 0.408 for Bejís to Manavgat and 0.470 for Manavgat to Bejís, against 0.406 and 0.467 in Table S16. Across the four λ values, Bejís to Manavgat gives 0.408 [0.389, 0.427], 0.410, 0.367 and 0.313, which are below chance at every λ. Manavgat to Bejís gives 0.470 [0.444, 0.494], 0.474, 0.466 and 0.457. No adapted direction of this pair has an interval above chance at any λ, so no conclusion depends on the

choice of λ. The third implementation, of Section S1.23 (R8c), reproduces the pipeline's CORAL matrix to within 0.009.

S1.3 Blocking scale

The transfer intervals were recomputed at 10-cell (≈5 km) blocking from the re-frozen per-cell predictions. The intervals became wider. The paired-delta verdicts changed from twelve positive, seven negative and one uncertain at 1 km to six, five and nine at 5 km. Coarser blocking therefore removed support from eight verdicts and added none. Across five bootstrap seeds, the 5 km counts were 5 to 6 positive, 4 to 5 negative and 9 to 11 uncertain, so they are not exact (`paper/labelfix_rerun/round5/out_official/transfer_ci_blocksize.csv`, `paper/labelfix_rerun/round6/seed_stability.json`).

The point estimates do not change, because the block size only sets the bootstrap resampling unit. Each point estimate is computed once over all target cells. Every verdict that changed moved to "no verdict", and no direction crossed the chance line. The counts of supported directions are therefore the least stable numbers in this paper.

Table 2 note (resampling units). The bootstrap resamples spatial blocks, so the reliability of an interval depends on the number of blocks with at least one burned cell. This number falls from 192 to 843 at 2 cells to 6 to 33 at 20 cells. An interval built on six such blocks has no useful coverage, so the 20-cell row should be read as indicative. The 10-cell row is the coarsest blocking that this design supports. There, every region has 16 to 70 blocks with burned cells, and the gain holds in all five regions.

S1.4 Predictor-window closure

The predictor window was closed 7 and 14 days earlier in all five regions. The thermal contribution stayed positive, with bootstrap support, everywhere. The direction of the change

differed between regions. In Manavgat, on the corrected label, it was +0.065 [+0.059, +0.072] at the standard window, +0.082 [+0.075, +0.090] at 7 days and +0.058 [+0.051, +0.065] at 14 days, all on the common cohort of that analysis (`paper/labelfix_rerun/exports/window_closure_manavgat_2021/`). It became stronger in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It was flat in Montiferru. It became weaker in Evia, from 0.156 to 0.149 to 0.135. The four regions other than Manavgat are not affected by the label correction (`paper/labelfix_rerun/exports/window_closure_<region>/tables/thermal_contributions.csv`). The gain therefore survives in every region, but it does not always improve.

S1.5 Quality screening of the coarse thermal input

The MODIS inputs were quality-screened in two regions and not in the other three. This split follows the export date and was never a design choice, but it causes a change at the input that is correlated with elevation (r = +0.615 in Manavgat). The Manavgat chain was therefore rebuilt from a quality-screened input on the corrected label and compared with the unscreened version. Because the current pipeline step7 refuses the unscreened raster (it has no nodata tag and 8.1 % exact zeros), the unscreened version uses the step7 of export time and the screened version the current one, so the two versions differ in code as well as in screening. The result did not change: elevation stayed at 0.232, no other signed AUC moved by more than 0.005 (downscaled LST, −0.0044), and the within-region gain moved from +0.067 to +0.068 (`paper/labelfix_rerun/round6/qc/qc_compare_manavgat_corrected.json`; Section S3.5(xii)). Elevation comes from the DEM and is not affected by the screening. The MODIS-based surface is used for only 2.14 percentage points of coverage. The Muğla input was not re-examined.

S1.6 Normalised against absolute dryness channels

Section 1.2 states that an internally normalised index should be less affected than raw land surface temperature by temperature differences between regions. The thermal set contains both kinds of channels, so this was tested directly by comparing feature sets.

In the two Muğla fires (Section S1.13), the normalised channels keep their direction and the absolute channels do not. This holds for one region across two fires. It does not hold across regions: `lst_anomaly_mean` is one of the predictors with a bootstrap-supported reversal on the original study areas (Table S10). For this reason, Section S1.14 removes it together with elevation. On this evidence, normalisation protects a channel against a difference between two seasons in one place, but not against a change of place.

Three feature sets were run over all twenty directions and all five within-region folds. The classifier, population, folds and bootstrap were kept fixed. The code stops unless its reference configuration reproduces the re-frozen exports, and it does: the largest absolute difference from the step9b transfer AUCs is 0.000000 across all twenty directions. Sources for this section and the next two: `paper/labelfix_rerun/round3/no_coord_channels.json` and `paper/labelfix_rerun/code/model_capacity.json`, summarised by `paper/labelfix_rerun/round7/r7e_supplement_tables.py`.

Table S23. Feature-set arms. Mean over the twenty transfer directions and the five regions.

Feature set	Mean transfer AUC	Directions > 0.5	Mean within-region AUC
Baseline only	0.519	13	0.797
Baseline + normalised anomalies	0.524	14	0.862
Baseline + absolute surface state	0.531	13	0.864
All ten features (reference)	0.527	13	0.896

The expected advantage was not found. Per direction, the normalised set minus the absolute set has a mean of −0.006. The difference is positive in 11 of 20 directions and ranges from

−0.099 to +0.087, so the spread is much larger than the mean difference. Within regions the two sets are almost equal, at 0.862 and 0.864. Each recovers most of the gap between the baseline (0.797) and the full model (0.896). The comparison uses two parts of one thermal set on one cohort. It does not test normalised dryness indices in general.

S1.7 The coordinate-informed channels

`downscaled_lst_mean` and `fused_lst_mean` come from a per-region downscaling model that uses coordinates as inputs (Section S3.4). This is the only way in which coordinate information enters the feature set. A surface smoothed by coordinates is informative locally and does not transfer. If the within-region gain depended on it, the gain would come from the smoothing, and the thermal state would add nothing. Both channels were therefore removed, and all models were refitted over all twenty directions and all five within-region folds.

Table S24. The within-region increment without the two coordinate-informed channels.

Region	Increment, full set	Without the two channels	Retained
Manavgat 2021	+0.067	+0.064	95 %
Bejís 2022	+0.056	+0.046	82 %
Muğla 2021	+0.116	+0.097	84 %
North Evia 2021	+0.153	+0.145	94 %
Montiferru 2021	+0.101	+0.105	103 %
Mean	**+0.099**	**+0.091**	**92 %**

The gain does not depend on these channels. It stays positive in every region and keeps 82 % to 103 % of its size. Mean transfer also does not change: 0.528 without the two channels, 0.527 with them, and 0.519 for the baseline alone.

S1.8 Model capacity

All numbers in the paper come from one random forest with unlimited depth and `min_samples_leaf = 3`. This configuration fits local structure well and extrapolates poorly.

Three other estimators were therefore run over the same twenty directions and five within-region folds.

Table S25. Four estimators, within region and in transfer.

Estimator	Within-region AUC	Within increment	Transfer AUC	Transfer increment	Above chance
Random forest, depth unlimited, leaf 3	0.896	+0.099	0.527	+0.007	13 of 20
Random forest, depth 6, leaf 50	0.838	+0.053	0.527	−0.011	13 of 20
Random forest, leaf 200	0.814	+0.046	0.522	−0.019	13 of 20
Penalised logistic regression	0.759	+0.046	0.481	−0.021	11 of 20

Regularisation does not improve transfer. The four estimators range from 0.481 to 0.527, with eleven to thirteen of twenty directions above chance. The linear model, which is designed to extrapolate, transfers worst. Within-region skill falls from 0.896 to 0.759 as capacity is reduced, but transfer does not rise. The thermal contribution to transfer is +0.007 for the standard forest and negative for all three alternatives. The estimators are not ranked: their pair-cluster intervals overlap, and every interval of the thermal contribution includes zero (Table S35, R8l).

S1.9 The four evaluations of Section 3.3, in full

Section 3.3 reports four evaluations of the same models (main-text Table 3). The values per scar, the controls and the thermal gain per scar are given here.

Table S1. The four evaluations, per scar. Natural-vegetation population, 5 km blocking, 2 km collar. Bejís and Manavgat have no row C, because each has only one burned area, and withholding it leaves nothing to train on. Source `paper/labelfix_rerun/code/matched_holdout.json`.

Region	Scar	Cells	Burned fraction	A whole region	B scar area	C scar withheld	D foreign
Manavgat 2021	1	3,965	0.74	0.882	0.743	n/a	0.542

Region	Scar	Cells	Burned fraction	A whole region	B scar area	C scar withheld	D foreign
Bejís 2022	1	1,657	0.66	0.824	0.575	n/a	0.589
Muğla 2021	6	1,266	0.72	0.777	0.564	0.595	0.571
Muğla 2021	1	1,244	0.59	0.777	0.654	0.561	0.514
Muğla 2021	10	940	0.68	0.777	0.674	0.617	0.634
Muğla 2021	8	954	0.57	0.777	0.761	0.542	0.623
North Evia 2021	1	3,059	0.87	0.864	0.745	0.465	0.564
Montiferru 2021	1	758	0.58	0.720	0.622	0.584	0.489
Montiferru 2021	5	195	0.34	0.720	0.461	0.458	0.475

S1.9.1 The pool controls

The same predictions were scored on a random sample of region cells with the burned fraction of the scar area. Over nine scars this gives 0.793, against 0.791 for the whole region. Matching the prevalence therefore changes nothing (−0.002 [−0.005, +0.001]). This is expected, because ROC-AUC does not depend on class balance, so it is a check of the code. Scoring the predictions on the scar area gives 0.644, a fall of +0.149 [+0.087, +0.211] from the prevalence-matched score. This replaces both the burned and the unburned cells, so each group was also replaced alone. The cost is always the prevalence-matched score minus the new score. Replacing only the unburned cells costs 0.139 [0.098, 0.181] (0.046 to 0.250 per scar). Replacing only the burned cells costs −0.002 [−0.055, +0.051] on average (−0.121 to +0.137 per scar). The cost therefore comes from the unburned cells. All unburned cells in a scar collar are next to the fire and share its terrain, land cover and weather. The unburned cells of a whole region also include easy distant cells (`paper/labelfix_rerun/code/pool_decomposition.json`, `prevalence_control.json`).

S1.9.2 The seven scars and the resampling unit

Rows A to D of Table 3 are means over the same seven scars: four in Muğla, two in Montiferru and one in Evia. Under the corrected label, the 2,151 added Manavgat burned cells join its one existing scar (2,934 of 2,935 cells). For this reason the controls above use nine scars and report 0.791 and 0.644, against 0.773 and 0.640 in Table 3. The intervals are

Student *t* intervals over the scars. Row A is a region-level value that is repeated for each scar of a region, so its interval is too narrow and should not be read as coverage. The same applies to A − B and A − C. With the region as the unit, A − B is +0.137 [+0.048, +0.226] over three regions and +0.160 [+0.090, +0.230] over all nine scars in five regions (`paper/labelfix_rerun/round7/round7_summary.json`). A − C with the region as the unit is +0.266 [−0.022, +0.553]. This interval is 3.6 times wider than the scar-level interval and includes zero. With the region as the unit, only the frame cost is established.

The scar definition does not drive the result. Over a range of minimum scar sizes and buffers, row C ranges from 0.541 to 0.561 on nine to six scars. Its means at 2, 5 and 10 km buffers are 0.546 (seven scars), 0.533 (seven) and 0.548 (six) (`paper/labelfix_rerun/code/scar_definition_sweep.csv`, `scar_control.json`). For single scars, row D varies strongly between sources (Table S4), but little after averaging. Row D uses 28 of the 36 source and scar combinations, and nine of them are below chance.

S1.9.3 Controls and the thermal-increment ladder, full specification

Three controls reuse the predictions of row A. Each is averaged over 20 draws without replacement. The *prevalence-matched* control draws $|H_K^+|$ cells from P_R and $|H_K^-|$ cells from $V_R \setminus P_R$. The *negative-pool* control keeps the drawn burned cells but uses the unburned cells of the scar, H_K^-. The positive-pool control does the opposite. A *within-region half-split* completes the set: the modelled cells are cut at the median of a grid axis, in both axes and both directions, and the transfer protocol of Section 2.8 is applied. A split is dropped when one half has only one class. The numbers of burned cells in the two halves are unequal (Table S2).

The thermal gain is evaluated with rows A to D on the seven scars of Table 3. Table S26 gives the values per scar at both block sizes. Over the seven scars, the region-wide gain is

+0.095 at 5 km blocking and +0.117 at 1 km. On the scar area, the same predictions give +0.021 and +0.056. Leave-one-scar-out gives +0.024, and a foreign-region model +0.008, at both block sizes. Blocked minus leave-one-scar-out is −0.004 [−0.070, +0.063] at 5 km and +0.031 [−0.027, +0.090] at 1 km. Region-wide minus scar area at 5 km is +0.074 [−0.009, +0.157] at scar level. With only three regions, no reliable region-level interval can be given for this difference. The within-region half-split gives +0.028, positive in 13 of 18 splits (`paper/labelfix_rerun/inference/ladder_summary.json`, `paper/labelfix_rerun/code/positive_control.json`).

Table S26. The thermal gain by scar: region-wide, on the scar area, withheld and foreign. Thermal minus baseline ROC-AUC. Region-wide and scar-area values at 5 km blocking, with 1 km in brackets. Leave-one-scar-out and foreign values do not depend on the blocking.

Region	Scar	Region-wide	Scar frame, same blocked model	Scar withheld	Foreign
Muğla 2021	6	+0.079 (+0.116)	−0.016 (+0.018)	+0.025	+0.033
Muğla 2021	1	+0.079 (+0.116)	+0.049 (+0.071)	−0.044	+0.014
Muğla 2021	10	+0.079 (+0.116)	+0.063 (+0.107)	+0.045	+0.042
Muğla 2021	8	+0.079 (+0.116)	+0.083 (+0.092)	+0.012	+0.037
North Evia 2021	1	+0.148 (+0.153)	+0.060 (+0.065)	+0.143	−0.038
Montiferru 2021	1	+0.100 (+0.101)	+0.066 (+0.086)	+0.060	−0.008
Montiferru 2021	5	+0.100 (+0.101)	−0.160 (−0.049)	−0.070	−0.023
Mean		**+0.095 (+0.117)**	**+0.021 (+0.056)**	**+0.024**	**+0.008**

Table S2. Within-region half-split, every split. The numbers of burned cells in the source and target halves are given because they are unequal. This is the main limit of this analysis, because a straight cut does not give two comparable halves. Two splits cannot be used, because one half of Manavgat has no burned cells. Source `paper/labelfix_rerun/code/positive_control.json`; checked row by row by `paper/code/appendix_tables.py`.

Region	Axis	Direction	Source positives	Target positives	Thermal AUC	Baseline AUC
manavgat 2021	east-west	low to high	1,795	1,140	0.761	0.793
manavgat 2021	east-west	high to low	1,140	1,795	0.712	0.638
manavgat 2021	north-south	low to high	0	2,935	n/a	n/a
manavgat 2021	north-south	high to low	2,935	0	n/a	n/a
bejis 2022	east-west	low to high	640	460	0.750	0.784
bejis 2022	east-west	high to low	460	640	0.578	0.560
bejis 2022	north-south	low to high	357	743	0.713	0.662
bejis 2022	north-south	high to low	743	357	0.603	0.613
mugla 2021	east-west	low to high	1,449	1,462	0.448	0.378
mugla 2021	east-west	high to low	1,462	1,449	0.636	0.628
mugla 2021	north-south	low to high	774	2,137	0.533	0.503
mugla 2021	north-south	high to low	2,137	774	0.294	0.549
evia 2021 extended	east-west	low to high	2,052	612	0.589	0.506
evia 2021 extended	east-west	high to low	612	2,052	0.645	0.482
evia 2021 extended	north-south	low to high	2,564	100	0.649	0.513
evia 2021 extended	north-south	high to low	100	2,564	0.533	0.495
montiferru 2021	east-west	low to high	410	129	0.475	0.461
montiferru 2021	east-west	high to low	129	410	0.507	0.513
montiferru 2021	north-south	low to high	438	101	0.583	0.511
montiferru 2021	north-south	high to low	101	438	0.536	0.447

Table S3. Leave-one-scar-out at a 2 km buffer, every scar. Only Muğla and Montiferru contain more than one burned area of at least 50 cells. Only in Muğla does every held-out case still leave a well-trained source model. The two Montiferru areas are very unequal, so one case keeps 472 burned source cells and the other 97. The mean is 0.579 for the four Muğla cases and 0.502 for the three others. The pooled mean, 0.546, is row C of Table 3.

Source `paper/labelfix_rerun/code/scar_control.json`; checked row by row by `paper/code/appendix_tables.py`.

Region	Component	Source positives left	Target positives	Target cells	AUC
evia 2021 extended	1	11	2,653	3,059	0.465
montiferru 2021	1	97	442	758	0.584
montiferru 2021	5	472	67	195	0.458
mugla 2021	6	1,997	914	1,266	0.595
mugla 2021	1	2,173	738	1,244	0.561
mugla 2021	10	2,272	639	940	0.617
mugla 2021	8	2,363	548	954	0.542

The evaluation populations of these analyses cannot be compared with each other or with the transfer targets. A held-out scar with its 2 km collar contains only unburned cells next to the fire, while a whole target region also contains easy distant cells. The burned fractions (34 to 87 %, against 7.0 to 28.7 %) follow from this difference and do not cause it.

Table S4. The foreign-region evaluation, by source. Each held-out scar area is scored with a model fitted on each of the other four regions. Row D of Table 3 is the mean over the seven scars that have a row C (28 of the 36 combinations below). Over all nine scars the mean is 0.556. Source `paper/labelfix_rerun/code/d_per_source.json`; checked row by row by `paper/code/appendix_tables.py`.

Target region	Scar	Mean over sources	Min	Max	Spread
Bejís 2022	1	0.589	0.519	0.631	0.112
North Evia 2021	1	0.564	0.374	0.704	0.329
Manavgat 2021	1	0.542	0.438	0.665	0.227
Montiferru 2021	1	0.489	0.457	0.513	0.056
Montiferru 2021	5	0.475	0.399	0.597	0.197
Muğla 2021	1	0.514	0.410	0.608	0.198
Muğla 2021	6	0.571	0.545	0.597	0.052
Muğla 2021	8	0.623	0.533	0.724	0.192
Muğla 2021	10	0.634	0.433	0.716	0.283

Over all 36 combinations the mean is 0.556, the range is 0.374 to 0.724, and ten combinations are below chance. The mean spread across the four sources for one scar is 0.183 over nine

scars and 0.187 over seven. Row D is averaged over sources so that it can be compared with row C, which is fitted on one region. For a single scar, the result still depends on which foreign region is the source.

Table S5. Prevalence is not the cause of the evaluation-area effect. The same fitted model and the same out-of-fold predictions are scored in three ways: on the whole region, on a random sample of region cells with the burned fraction of the scar area, and on the scar area. Twenty draws per scar, seed 42. Nine scars are used, because this control needs no leave-one-scar-out model. Values at 4 dp, as stored in the source `paper/labelfix_rerun/code/prevalence_control.json`; checked row by row by `paper/code/appendix_tables.py`.

Target region	Scar	Burned fraction	A whole region	A′ prevalence-matched	B scar area
Manavgat 2021	1	0.74	0.8822	0.8837	0.7434
Bejís 2022	1	0.66	0.8245	0.8251	0.5748
Muğla 2021	6	0.72	0.7773	0.7782	0.5640
Muğla 2021	1	0.59	0.7773	0.7787	0.6540
Muğla 2021	10	0.68	0.7773	0.7835	0.6735
Muğla 2021	8	0.57	0.7773	0.7796	0.7609
North Evia 2021	1	0.87	0.8642	0.8632	0.7451
Montiferru 2021	1	0.58	0.7199	0.7179	0.6215
Montiferru 2021	5	0.34	0.7199	0.7285	0.4611
Mean			**0.7911**	**0.7932**	**0.6443**

A minus A′, the effect of prevalence alone, is −0.002 [−0.005, +0.001]. A′ minus B, which replaces both groups of cells, is +0.149 [+0.087, +0.211]. Section S1.9.1 shows that it comes from the unburned cells.

S1.10 The transfer-gap decomposition, in full

Among the directions that started below chance, the recovered fraction of Section 2.10 is at most +0.28 (Bejís to Evia). Seven directions have negative recovery, five of them with intervals fully below zero. Manavgat to Muğla (−0.03 [−0.06, +0.01]) and Muğla to Bejís

(−0.07 [−0.16, +0.01]) are negative at the point estimate only. Values are from `four_aoi_decomposition.csv` of the re-frozen outputs (`paper/labelfix_rerun/round5/tables/corrected/`). The within-region reference in the denominator is not matched to a transfer evaluation (Section 3.3), and the raw column depends on the study area (Section 3.4). The fractions are therefore only valid within this protocol.

Table S6. Transfer-gap decomposition (four-AOI set, 12 directions). Within = within-region thermal AUC of the target. Best adapted = the better of z-score and CORAL; this choice uses target labels. Recovered fraction = (adapted − raw)/(within − raw), signed and not clipped, with a paired bootstrap CI (1000 replicates, 2-cell blocks). Montiferru directions are not part of this decomposition. The status column shows whether the *adapted* value is above chance, using the 2-cell adapted intervals of Table S16.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	adapted above chance
Muğla→Manavgat	0.908	0.345	0.485 (z-score)	+0.25 [+0.22, +0.28]	adapted, chance not excluded
Evia→Bejís	0.918	0.448	0.549 (z-score)	+0.22 [+0.16, +0.27]	adapted above chance
Bejís→Manavgat	0.908	0.314	0.406 (CORAL)	+0.15 [+0.13, +0.18]	adapted still below chance
Manavgat→Bejís	0.918	0.396	0.467 (CORAL)	+0.14 [+0.09, +0.17]	adapted still below chance
Manavgat→Muğla	0.859	0.438	0.427 (z-score)	−0.03 [−0.06, +0.01]	**negative recovery**
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	**negative recovery**
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	**negative recovery**
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	**negative recovery**
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	**negative recovery**
Manavgat→Evia	0.912	0.654	0.529 (z-score)	−0.48 [−0.56, −0.42]	**negative recovery**
Evia→Manavgat	0.908	0.677	0.417 (CORAL)	−1.13 [−1.27, −1.00]	**negative recovery**

S1.11 The thermal sign, stratified

Section 3.4 reports that four of five regions show a negative association between pre-fire surface temperature and burning, and that Manavgat shows the same once elevation is held. The values per region are given here. Signed AUC against `burned` within the 10 km collar. The stratified columns pool the concordance within deciles of the named variable, and each decile is weighted by its number of burned and unburned cell pairs. Sources: `paper/labelfix_rerun/code/matched_frame_gap.csv` (LST raw, within NDVI, within distance; NDVI columns), which can be recomputed by `paper/code/verify_matched_gap.py`, and `paper/labelfix_rerun/round7/r7b_lst_given_terrain.csv` (LST within elevation, and LST detrended on elevation).

Table S27. The LST association, stratified, 10 km collar. Point estimates; 10-cell block-bootstrap intervals for every cell are in `paper/labelfix_rerun/round8/r8e_stratified_intervals.csv`, and the Manavgat values are summarised in Table S35.

Region	LST raw	LST within elevation	LST detrended on elevation	LST within NDVI	LST within distance	NDVI raw	NDVI within LST
Manavgat	0.522	**0.403**	**0.409**	0.556	0.454	0.551	0.550
Bejís	0.405	0.509	0.411	0.486	0.350	0.618	0.574
Muğla	0.332	0.363	0.365	0.404	0.367	0.652	0.547
Evia	0.286	0.327	0.312	0.279	0.319	0.663	**0.380**
Montiferru	0.376	0.392	0.416	0.368	0.485	0.582	**0.405**

Three results follow. First, the raw LST association of Manavgat is above 0.5, and it falls well below 0.5 once elevation is held. This holds both for deciles and for removing the linear dependence of LST on elevation in the region (−5.6 K per km). On the full study area the same holds: 0.665 raw, against 0.455 and 0.446. The positive signal in Manavgat therefore comes from terrain. With elevation held, Manavgat has the same sign as the other four regions

at the point estimates; the interval excludes 0.5 on the collar but not on the original study area (Table S35). Second, in the other four regions the sign stays below 0.5 within elevation deciles in three regions and within NDVI deciles in all four. In Bejís it moves to 0.509 within elevation deciles and stays below 0.5 after detrending. Third, NDVI reverses in two regions once LST is held, but LST reverses in none of the four once NDVI is held. Holding northness (the cosine of aspect) through deciles does not change the sign of the LST association in any region (Table S35).

S1.12 The LST anomaly under the difference instrument

Section 3.4 reports two elevation reversals on the collar, both involving Manavgat, that meet the per-comparison criterion. The LST anomaly differs between regions under a weaker test: the two regions are bootstrapped separately and their difference is taken. The three pairs below have point estimates on opposite sides of 0.5 and a difference interval that excludes zero, all for `lst_anomaly_mean`. Signed AUC within the 10 km collar; the two regions are bootstrapped separately with 10-cell spatial blocks, 1000 replicates, seed 42.

Table S7. Between-region differences in the signed LST-anomaly association, 10 km collar. Pairs with point estimates on opposite sides of 0.5 and a difference interval that excludes zero. Source `paper/labelfix_rerun/inference/reversal_family_holm.csv` (collar frame); checked row by row by `paper/code/appendix_tables.py`.

Pair	AUC A	AUC B	Difference	95 % CI
Bejís vs Evia	0.392	0.584	−0.191	[−0.295, −0.083]
Evia vs Montiferru	0.584	0.400	+0.184	[+0.012, +0.321]
Bejís vs Muğla	0.392	0.507	−0.115	[−0.219, −0.002]

On the collar, 27 of the 90 feature-by-pair differences have intervals that exclude zero, against about 4.5 expected by chance at 5 %. Four of the 27 are for this feature. Three of these four have point estimates on opposite sides of 0.5, and two of the three involve Evia. With a Holm correction on normal-approximation p-values, 12 of the 90 remain. With the stricter

intersection-union form, which also requires the association in each region to be significant, none remains (Section S1.19). The nine features carry about two independent thermal signals (Section 2.4), so the 90 comparisons are not independent. This result is weaker than a reversal.

S1.13 The same-geography event pair

This comparison is reported only here: the two populations overlap, only elevation and slope can be evaluated, and the result does not survive the frame test (below). The design and two differences from the twenty-direction matrix are given here. The protocol is in Section S3.3.

This comparison was designed to keep the place fixed and change only the fire. Muğla burned in 2021 and again eleven months later, on the same grid and with the same processing. The 2022 population is the 2021 population without the 2021 scar: 41,730 rows with 2,911 burned for 2021, against 38,790 rows with 331 burned for 2022. The 2021 population has 70 blocks of 5 km with burned cells, and the 2022 population has 11. Eleven is below the minimum of sixteen used in this design, so the 2022 intervals are indicative, like the 20-cell row of Table 2. Values for all features are in Table S12.

On the original study area, elevation reverses with bootstrap support. In 2021, higher ground burned more, at 0.611 [0.532, 0.690]. In 2022, lower ground burned more, at 0.296 [0.230, 0.355]. The intervals do not overlap, and the difference is −0.317 [−0.414, −0.220].

The collar removes this reversal. The 2022 fire is one compact scar inside the whole Muğla rectangle, so 93.2 % of its cells lie more than 10 km from any burned cell (median 43.6 km), against 55.3 % in 2021. This is the largest share of distant cells in the cohort. On the 10 km collar, the 2021 value hardly changes (0.611 to 0.606, still supported). The 2022 value moves from 0.297 to 0.565, to the same side of 0.5 as in 2021, with an interval that includes chance. Slope does not reverse on either frame.

Only elevation and slope can be evaluated. The modelling export of the 2022 fire is not in this repository. The released script (`paper/code/verify_mugla_collar.py`) rebuilds the 2022 population from the 2021 predictor file with the 2022 burned mask. This is valid for elevation, slope and land cover, which are the same in both years, but not for the seasonal channels. Compared with the pipeline's own 2022 export, the rebuilt data agree on elevation (0.297 against 0.296) and slope (0.559 against 0.558). They differ on every seasonal channel: by +0.136 on `lst_anomaly_mean`, +0.092 on `tvdi_difference_mean`, −0.078 on `ndvi_mean` and −0.025 on `current_tvdi_mean`. Only elevation and slope are therefore given a collar result here.

Feature	2021, full	2022, full	full verdict	2021, collar	2022, collar	collar verdict
elevation_mean	0.611 [0.529, 0.692]	0.297 [0.229, 0.363]	**supported reversal**	0.606 [0.525, 0.685]	0.565 [0.450, 0.677]	none
slope_mean	0.637 [0.584, 0.689]	0.559 [0.463, 0.643]	none	0.635 [0.578, 0.692]	0.457 [0.356, 0.548]	none
the state-dependent channels	n/a	n/a	n/a	n/a	n/a	**not evaluable here**

Source `paper/labelfix_rerun/code/mugla_two_event_collar.csv`. The two populations are not separate samples. They share 38,789 of the 38,790 cells of the 2022 population. Elevation, slope and land cover are identical in all 73,098 grid cells, so only NDVI and the six thermal channels differ between them. The 2021 burned cells were also removed from the 2022 population. In the 2022-to-2021 direction, therefore, no burned target cell is in the source training data, while 38,789 of the 38,819 unburned target cells are. Membership of the training data alone separates the two classes of the 2021 target at ROC-AUC 0.9996. The effect of this on a transfer estimate between the two years cannot be bounded. The known part of the bias makes the reversal smaller: the removed cells are high (median 563 m), so removing them takes high unburned cells out of the 2022 population.

S1.14 The two interventions, in full

(a) Pooled multi-region training (Fig. 6). A model trained on the pooled populations of the other four regions is better than the mean of the four single-source models for one target of five, Evia, at 0.715 [0.668, 0.757] against a mean of 0.569. For the other four targets it is worse than this mean, by 0.009 to 0.057. It is better than the best single source only for Evia (0.654). The best source can only be chosen with target labels, so it is only a reference. For every target the pooled model stays 0.20 to 0.48 AUC below the within-region ceiling. For Manavgat and Bejís it is below chance, and only for Manavgat with interval support, at 0.426 [0.369, 0.486]. Outside Evia, pooling does not recover what single-source transfer loses (`paper/labelfix_rerun/code/loro_all.json`, `paper/labelfix_rerun/round7/r7d_pooled_vs_single.csv`).

(b) Removing the direction-reversing features. Under the original label, two predictors reversed between regions with bootstrap support on the original study areas: `elevation_mean` and `lst_anomaly_mean`. This analysis keeps these two and does not re-select under the corrected label (Table S10). Section 3.4 shows that the study area decides much of the support, so this analysis measures the cost of removal under the original protocol. Without the two predictors, mean within-region AUC falls by 0.076, with support in every region (−0.035, −0.130, −0.073, −0.063 and −0.079 for Manavgat, Bejís, Muğla, Evia and Montiferru; every interval fully below zero). Mean transfer changes by +0.014 [−0.028, +0.056] (Student *t* over the ten region pairs, as plotted in Fig. 7), which includes zero; the primary pair-cluster bootstrap gives [−0.018, +0.049] (`paper/labelfix_rerun/round8/r8q_summary.json`) (source `paper/labelfix_rerun/round3/feature_drop_transfer.json`, summarised by `paper/labelfix_rerun/round7/r7e_supplement_tables.py`).

Two points qualify this cost. First, most of it does not come from the thermal set. Removing elevation alone lowers AUC by 0.056 (0.896 to 0.840). Removing the LST anomaly alone lowers it by 0.013 (0.896 to 0.883). About three quarters of the cost therefore comes from removing a baseline terrain variable. Second, both predictors were chosen because they reverse, on the same data on which the costs are estimated, and no correction is applied.

Table S28. Feature-removal configurations.

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.896	0.527
drop `elevation_mean`	0.840	0.533
drop `lst_anomaly_mean`	0.883	0.529
drop both	0.820	0.541

Fig. 7 shows the four configurations together. A local cost is measured, but no transfer gain. This is not an exchange, because neither transfer effect is established: the thermal contribution to transfer is +0.007 and removing the reversing predictors gives +0.014, and both intervals include zero.

S1.15 The paired thermal contribution to transfer

The signed associations on both frames are given in Tables S9, S10 and S14, and their stratified values in Section S1.11; Section 3.4 summarises them. This section gives the uncertainty of the paired thermal contribution to transfer.

The thermal and baseline matrices were differenced direction by direction. The mean is +0.007. The single contributions range from −0.148 to +0.132, with twelve positive and eight negative, so a mean near zero comes from positive and negative values that cancel each other. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on the resampling unit. All four units in Table S29 give intervals that include zero (1000 replicates where resampled; source `paper/labelfix_rerun/code/transfer_delta_ci.json`):

Table S29. The mean paired thermal contribution to transfer, by resampling unit.

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive bootstrap	20	[−0.023, +0.037]
Unordered pairs, cluster bootstrap (primary)	10	[−0.020, +0.038]
Unordered pairs, Student *t* on pair means	10	[−0.028, +0.043]
Regions, leave-one-out jackknife	5	[−0.018, +0.033]

Three further units can be computed on this frame (20,000 replicates; `paper/labelfix_rerun/round8/r8d_summary.json`): clustering by target region gives [−0.006, +0.022], clustering by source region [−0.008, +0.026] and the pigeonhole bootstrap [−0.028, +0.045]. Leaving out the naive direction bootstrap, which ignores the dependence between directions, six units can be computed, and all seven intervals include zero.

Leaving out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0050, +0.0072, +0.0010 and +0.0087. No single region therefore carries the mean or changes its sign. These units resample only between directions. They do not include the sampling variability within a direction.

S1.16 The contrast pair, in full

Manavgat and Muğla are in the same country, burned in the same year, and are 306 km apart. Their burned cells have the most similar environmental range of any pair in the matrix. Bejís and Montiferru have the least similar range (Fig. 8). Values for both pairs, on both frames, are in Table S18.

For Manavgat and Muğla on the original study areas, seven of nine feature-response directions point opposite ways, and all six features supported in both regions have opposite signs, including elevation. On the collar the elevation reversal remains, at 0.376 [0.300, 0.465] in Manavgat against 0.606 [0.525, 0.685] in Muğla. Transfer is below chance in both directions on both frames: 0.438 and 0.345 as drawn, 0.493 and 0.433 under the collar. On the collar, the interval of Muğla to Manavgat, [0.381, 0.491], excludes chance. The most similar

pair is among the weakest in the matrix, but it is not the weakest. On the collar, the weakest direction is Manavgat to Bejís at 0.407 and the strongest is Muğla to Evia at 0.728. The niche-overlap and applicability measures of Table S18 are computed on the original study areas.

Bejís and Montiferru are at the other extreme. Their burned ranges hardly overlap, and they have the most different values on every overlap measure. They transfer above chance in both directions, but only at the point estimate: neither direction has interval support at 5 km blocking. The applicability measure was not produced for Montiferru. At the point estimates, high niche overlap did not ensure transfer, and low overlap did not prevent it. This rests on two pairs. It is not a correlation across pairs; that is tested in Section S1.21.

S1.17 The sensitivity arms, summarised

Eight design choices were changed, one at a time, with everything else kept fixed: the Evia study area and its prevalence, the CORAL regularisation, the block size, the closing date of the predictor window, the quality screening of the coarse thermal input, normalised against absolute dryness channels, the removal of the coordinate-informed channels, and the capacity of the classifier (Sections S1.1 to S1.8). None changes a conclusion of the main text. Two of them affect how the results should be read.

The secondary population. Section 2.5 defines a secondary population of all valid cells, including cropland. It exists only for the within-region models in Manavgat and Bejís, so it is a two-region sensitivity analysis, and no transfer quantity is defined on it. Across the two regions and three block sizes, the thermal gain is +0.048 to +0.061, with every bootstrap interval above zero, against +0.045 to +0.067 in the primary population. The paired difference between the populations is −0.011 to +0.011, with no consistent sign (`paper/labelfix_rerun/pipeline/robustness/step8_large_block_primary_all_valid/`;

Table 2). The gain is therefore not caused by excluding cropland. This analysis says nothing about transfer.

S1.18 Distance within a region

This analysis measures how skill falls with distance inside one region. A model is fitted on one half of a region and applied to the other half. Target cells are grouped by their distance from the training cells (`paper/code/distance_curve.py`; corrected output `paper/labelfix_rerun/round3/distance_curve.json`, summarised in `paper/labelfix_rerun/round7/r7h_distance_curve.csv`). Means are unweighted over bins. The bins contain 1 to 1,067 burned cells and have no intervals.

Table S32. Within-region skill by separation from the training cells.

Separation from training cells	Bins	Mean target AUC
0 to 5 km	18	0.709
5 to 10 km	16	0.570
10 to 20 km	11	0.519
20 to 40 km	6	0.499
40 to 80 km	3	0.541
80 to 160 km	1	0.421
cross-region, 306 to 2,802 km	20	0.527

At 10 to 20 km inside one region, the model is close to chance. This limits how far such a model can be used from its training cells. It agrees with Section 3.3, where withholding a scar or using a foreign model had no measurable cost.

This does not show that the cross-region failure is only an effect of distance. Once the curve reaches chance, any cross-region mean near 0.5 must lie on its continuation, so this comparison cannot test it. Also, a model that carries no information does not give a reliably reversed ranking, but Manavgat to Bejís is below chance with interval support both on the original study area and on the 10 km collar.

Four limits apply. First, the two distance ranges do not overlap: within-region distances are 2 to 86 km, and cross-region distances start at 306 km. Second, the far bins contain few cells, so their means should not be read closely; the rise at 40 to 80 km is not evidence of an effect. Third, the near bins include the spatial autocorrelation that blocked validation is designed to remove, so 0.709 is an upper bound on near-field skill. Fourth, the design separates distance from crossing a study-area boundary, but not from the land cover, terrain and fire history that change with distance. The analysis gives a length scale for the fall of skill within a region; it does not identify a cause.

S1.19 The frame test, elaborated

Section 3.4 gives these results. Sources: `paper/labelfix_rerun/code/aoi_frame_auc.csv` (signed associations by frame), `paper/labelfix_rerun/code/collar_frame_bootstrap.csv` (their 10-cell intervals), `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv` (the transfer matrix by frame) and `paper/labelfix_rerun/code/matched_frame_gap.csv` (the matched reference); code in `paper/code/`.

Construction. Every region is restricted to cells within 10 km of any burned cell. This removes only distant unburned cells. Every burned cell is at distance zero and is kept at any radius, so radii were compared to guard against a favourable choice. The interval criterion was evaluated at the 10 km collar; the 5 and 10 km radii agree at the point estimate. The collar strongly reduces the number of cells in two regions, so part of any loss of support is a loss of power. On average, 2.3 features per direction are supported in both regions on the original study areas, and 2.2 on the collar.

Multiplicity. Each frame has a family of ninety feature-by-pair comparisons (nine features, ten region pairs). Table S30 gives how many pass each criterion. The per-comparison criterion of Section 3.10 leaves two supported reversals on the collar, both on elevation and

both involving Manavgat: Manavgat against Muğla, which is also supported on the original areas, and Manavgat against Evia, which is supported only on the collar. Manavgat against Bejís, supported on the original areas, misses support on the collar by 0.004. Neither survives the Holm intersection-union correction, which adjusts the tests of both regions together (adjusted $p = 0.33$ for Manavgat against Evia and 0.79 for Manavgat against Muğla, normal approximation).

Table S30. Reversal family, ninety feature-by-pair comparisons per frame. Source `paper/labelfix_rerun/inference/reversal_family_holm.csv`, `reversal_per_region.csv`.

Criterion	Full frame	10 km collar
Difference interval excludes zero	39	27
… with opposite-sided point estimates	31	21
Per-comparison criterion (each region's own interval excludes 0.5, opposite sides)	14	2
Holm, normal approximation, difference test	27	12
Holm intersection-union (both regions' tests)	5	0

Similarity measures on the collar. Two of the twenty similarity measures are built from the signed associations, and they were recomputed on the collar. The sign-agreement fraction over supported features is ρ = +0.52 [−0.27, +0.87] on the original study areas. On the collar it takes only the values 0 and 1 across sixteen defined directions, and it correlates with collar transfer at ρ = +0.38 [−0.17, +0.85]. The all-feature cosine is +0.44 [−0.29, +0.80] on the original study areas and correlates with collar transfer at +0.57 [+0.05, +0.88]. This is the only interval of a similarity measure in the study that excludes zero (`paper/labelfix_rerun/round6/diag_collar/`; intervals from `paper/labelfix_rerun/round7/r7g_collar_diagnostics.json`, which reproduces the original intervals exactly). It is not treated as a result for three reasons. It is one of about forty uncorrected tests. It rests on twenty directions that share regions. The collar is defined from

the burned cells, so the measure needs target labels twice. The other eighteen measures were not recomputed on the collar.

Data provenance. A quality-screening rebuild overwrote the 500 m modelling datasets of Manavgat and Muğla at the standard pipeline path after the frozen tables were computed. Each new file differs in `downscaled_lst_mean` and `fused_lst_mean`. The pipeline records a SHA-256 for the modelling dataset of each region, and this hash identifies the frozen copy. Every analysis of Section 3.4 reads each region through `paper/code/_canonical.py`, which checks the hash on load, and reads Manavgat from the corrected re-freeze (SHA-256 `5a5e876c…`). The provenance of the collar transfer matrix is in `paper/labelfix_rerun/round5/collar/PROVENANCE.md`.

Table S8. Cross-region transfer on comparable study areas. Natural-vegetation population, thermal model, twenty ordered directions per row. Above and below chance are point counts. The supported counts use a 10-cell (≈5 km) spatial-block bootstrap on the target, 1000 replicates, seed 42. Table S16 gives the original matrix under 2-cell (≈1 km) blocking. Bounds per direction are in `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`.

Source frame	Target frame	Mean target AUC	Above chance	Below chance	Supported above / below	Paired thermal delta
full	full	0.527	13 of 20	**7**	9 / **6**	+0.007
full	10 km	0.559	14 of 20	6	10 / 2	+0.008
10 km	full	0.546	14 of 20	6	10 / 5	+0.014
10 km	**10 km**	**0.589**	**15 of 20**	**5**	**13 / 2**	**+0.024**
5 km	5 km	0.591	16 of 20	4	11 / 0	+0.017

The baseline changes with the frame. On the 10 km collar, the static baseline transfers at 0.565 and the thermal model at 0.589, a paired difference of +0.024 instead of +0.007. The static baseline still does not transfer well. The number of directions below chance falls from seven to five at the point estimate, and from six to two with interval support. The largest rises

are Bejís to Evia (0.383 to 0.602) and Bejís to Manavgat (0.314 to 0.452). The only large fall is Evia to Manavgat (0.677 to 0.594).

The matched reference. The within-region reference was recomputed on the same frame at 5 km blocking (`paper/code/verify_matched_gap.py`):

Table S33. Within-region reference and transfer on matched frames.

Frame	Within-region (5 km blocking)	Mean transfer	Gap
full rectangle	0.814	0.527	0.287
10 km collar	**0.786**	**0.589**	**0.197**
5 km collar	0.759	0.591	0.169

Paired by target region, the shortfall on the collar is +0.197 [+0.091, +0.303] (Student *t* over the five target regions; +0.081 Montiferru, +0.319 Manavgat, +0.172 Muğla, +0.202 Bejís, +0.211 Evia). The shortfall remains on every matched row. It is 0.197 on the collar against 0.29 in the unmatched comparison, and it becomes smaller as the frame moves closer to the fire. On the 5 km collar, the within-region gain falls by half but stays positive in all five regions, with a mean of +0.042 (Manavgat +0.035).

S1.20 The transfer matrix and adaptation, elaborated

Section 3.5 gives these results.

Raw transfer. Raw target AUC ranges from 0.314 to 0.677 (Fig. 4). At 2-cell blocking, eleven of 20 directions are above chance with interval support and seven are below: both directions between Manavgat and Bejís, both between Manavgat and Muğla, both between Bejís and Evia, and Montiferru to Manavgat. At 10-cell blocking the same point estimates give 9 above, 6 below and 5 uncertain. Manavgat to Muğla loses its support, and no direction changes side of the chance line. Of the directions below chance, Manavgat to Bejís (0.407 [0.323, 0.493]) and Muğla to Manavgat (0.433 [0.381, 0.491]) keep interval support on the 10

km collar, but both intervals include chance on the 5 km collar. The lowest direction on the original study areas is Bejís to Manavgat, at 0.314 [0.296, 0.332].

Label-free adaptation. Under region-wise z-scoring the twenty directions range from 0.302 to 0.630, and under CORAL from 0.406 to 0.624. The change is not uniform: under z-scoring, Bejís to Manavgat falls further below chance, from 0.314 to 0.302. With the better of the two methods in each direction, 15 of the 20 directions end closer to chance and 5 end further from it. Four of these five involve Montiferru and move up. The fifth, Manavgat to Muğla, moves down from 0.438 to 0.427. Under CORAL alone, sixteen of twenty end closer to chance (`paper/labelfix_rerun/round5/matrix20_official.csv`). In the six directions where raw transfer was below chance, the best label-free method recovers at most 28 % of the gap to the within-region reference (Bejís to Evia). On the twelve directions of the decomposition, seven show negative recovery (Section S1.10).

Seed stability. Across five bootstrap seeds, every 2-cell verdict is stable. At 10 cells, one level verdict and two paired-delta verdicts change: the level verdict of Evia to Bejís, and the paired-delta verdicts of Manavgat to Muğla and Evia to Montiferru (`paper/labelfix_rerun/round6/seed_stability.json`). The random-forest seed is fixed, so all intervals per direction are conditional on one fitted source model.

Jackknife. Leaving out one region at a time moves the mean paired contribution between +0.001 (without Evia) and +0.015 (without Manavgat). Its sign never changes.

S1.21 The similarity diagnostics

Twenty similarity measures from five families were each rank-correlated with raw thermal transfer AUC over the directions on which they are defined. One pair-based bootstrap is used for all of them (unordered pairs resampled with replacement, both directions kept, 2000 replicates, seed 42). The measures were defined on 8 August 2026, together with their results

under the original label, and were not re-selected after the label correction. Table S17 lists all twenty. The source is `paper/labelfix_rerun/round5/out_official/all_diagnostics_vs_transfer.csv`, and the original-label values are in `paper/labelfix_rerun/round5/s6_diagnostics_20.md`.

On the original study areas, no interpretable measure has an interval that excludes zero. The largest correlations are for the conditional family: the sign-agreement fraction over supported features, at $\rho = +0.52$ [−0.27, +0.87], and its cosine, at +0.49 [−0.24, +0.87]. Under the original label these two were the only rows with intervals that excluded zero (+0.84 and +0.81). This did not hold after the label correction. The marginal family (applicability, dissimilarity, climatic distance, domain classification), the niche-overlap family and the regime family all include zero. Geographic distance does not order the matrix ($\rho = -0.17$ [−0.87, +0.84] on the twelve directions where it is defined). The domain classifier separates source from target at AUC ≥ 0.96 for every pair. Because it always succeeds, it gives no ordering. The twentieth measure, the vector Spearman of signed AUCs over supported features, is defined on only six directions. Its interval is degenerate (the upper bound equals the point estimate), so it is listed but not interpreted.

The collar. Section S1.19 gives the two measures recomputed on the collar. One of them, the all-feature cosine, orders collar transfer with an interval that excludes zero. It is reported with its limits and is not used.

Limits. Signed associations need burned labels in both regions, so the conditional family cannot be used before a fire. The marginal family can be used, but it fails. Ten independent region pairs give little power, and a moderate ordering would usually be missed. These results therefore show a failure to find an ordering; they do not prove that none exists (Section

S3.5(xiii)). Measures based on interval support are unstable when bounds lie near 0.5 (Section S3.5(viii)).

S1.22 Additional robustness analyses

The analyses in Table S21 were run after the main results. They test whether the main quantities depend on analysis choices. None of them was used to select a reported configuration.

Table S21. Additional robustness analyses (post hoc). Sources under `paper/labelfix_rerun/`: `geometry/` (a1 to a5), `inference/` (units, equivalence, ladder) and `round7/`.

Analysis	Question	Result
Edge-excluded frame cost (a1)	Is the frame cost carried by the scar's edge cells?	Over nine scars, excluding 0, 1 or 2 cells of the scar edge gives A − B = 0.147 [0.086, 0.208], 0.137 [0.069, 0.205] and 0.129 [0.055, 0.204]; with the region as unit 0.160, 0.150 and 0.145, every interval above zero
Placebo collars (a2)	Is the cost a property of any fire-shaped frame?	The scar's real positives scored against the negatives of a same-shaped collar placed away from any burned cell reproduce A (A − placebo −0.002 [−0.056, +0.052], eight scars with placements); the real collar costs +0.155 [+0.108, +0.202] more, so the cost is the fire-adjacent negatives
Frame cost against the share of distant cells (a3)	Is the cost larger where the study area has more distant cells?	No. Region-level slope 0.005 [−0.41, +0.42], r = 0.02 (n = 5); scar-level slope −0.049 [−0.34, +0.24] (n = 9, scars not independent). The test has little power
Label-free target frames (a4)	Does a frame drawn without target labels change transfer?	Trimming the target to the source's per-feature range, or to the source's area of applicability (Meyer and Pebesma 2021), gives means of 0.520 to 0.574, and a 20 km window 0.482; every paired thermal Δ interval spans zero
Metric dependence (a5)	Is the frame cost specific to ROC-AUC?	Region-wide minus scar-frame over nine scars: partial AUC (FPR ≤ 0.1) 0.088 [0.048, 0.128]; average precision −0.342 [−0.477, −0.206], higher on the scar frame because its prevalence is higher
Collar within-region increment	Does the within-region increment survive the collar?	+0.083 [+0.037, +0.129] over five regions (Student *t*), against +0.087 [+0.037, +0.136] as drawn
Equalised Δ by resampling unit	Does the collar Δ exclude zero?	Table S31: two of six computable units exclude zero; the two-way estimators are undefined

Analysis	Question	Result
Equivalence of the equalised Δ	How large a gain is excluded?	Within ±0.05 under four of six computable units, but not under the pigeonhole bootstrap or the region jackknife (90 % upper bounds 0.052 and 0.055); no gain above 0.047 at 90 % (pair cluster); not within ±0.02 under any unit. On the original study areas the gain is within ±0.05 under all six units
Within minus transfer increment	Is the local-versus-portable contrast itself supported?	Full frame +0.079 [+0.014, +0.145]; 10 km collar +0.059 [+0.007, +0.110]; 5 km collar +0.026 [−0.012, +0.063]
Reversal multiplicity	Do the collar reversals survive correction?	Table S30: two per-comparison reversals, none after Holm intersection-union
scikit-learn version	Do point estimates depend on the library version?	Under 1.5.2 against 1.9.0, single directions move by up to 0.047 and means by at most 0.003; one support count changes (`round7/sklearn152/compare_vs_1_9_0.json`)

Table S31. The equalised (10 km collar) paired contribution by resampling unit. 20,000 replicates where resampled; 95 % intervals. Source `paper/labelfix_rerun/inference/units.json`.

Unit	95 % interval	Excludes zero
Directions, naive	[−0.001, +0.049]	no
Unordered pairs, cluster bootstrap (primary)	[−0.003, +0.051]	no
Target region, cluster bootstrap	[+0.010, +0.040]	yes
Source region, cluster bootstrap	[+0.010, +0.042]	yes
Pigeonhole (source and target)	[−0.005, +0.060]	no
Two-way cluster (CGM)	undefined (negative variance)	n/a
Dyadic-robust	undefined (negative variance)	n/a
Leave-one-region-out jackknife, *t*	[−0.017, +0.065]	no

The primary unit is the pair cluster (Section 2.7), and on it the paired contribution on the collar includes zero. With five regions, a percentile bootstrap over regions has at most 126 distinct resamples, so the two region-clustered intervals are coarse.

S1.23 Analyses added after the pre-submission review

The analyses in Table S35 were run after an internal review of the manuscript. They test the main results against the fitted forest, the adaptation method, the inference, the land-cover map, the burned-area product, terrain aspect and a weather reference. Sources are in `paper/labelfix_rerun/round8/`, which also describes each script.

620 **Table S35. Analyses added after the pre-submission review (post hoc).**

Analysis	Question	Result
Random-forest seed (R8a)	Do the transfer results depend on the fitted forest?	With ten seeds, mean transfer on the original study areas ranged from 0.525 to 0.530 and the mean thermal gain from +0.005 to +0.008. The seed-to-seed standard deviation of a direction had a median of 0.005 (maximum 0.015), and no direction changed side of chance. On the 10 km collar, mean transfer ranged from 0.588 to 0.592, and one direction (Montiferru to Manavgat) changed side. Averaging the predictions of the ten forests gave 0.527 (gain +0.006) and 0.590 (gain +0.025). Seed 42 reproduces the published matrix exactly
CORAL variants and placebo (R8c)	Is the movement toward chance specific to aligning with the target?	CORAL with $\lambda = 1$ gave a mean of 0.512 and CORAL on two thermal principal components 0.523, against 0.527 without adaptation. A placebo that aligns the source with the covariance of a third region instead of the target gave 0.507 and moved 15 of 20 directions closer to chance, against 16 of 20 for CORAL. The movement is therefore not specific to the target. The published CORAL values were reproduced to within 0.009
Permutation test of the similarity measures (R8d)	Does any measure predict transfer under a permutation test?	No. With all 120 permutations of the five regions, the smallest two-sided p-value was 0.10 (vector Spearman, six directions only); the two supported-feature conditional measures gave 0.125 and 0.15, and all other measures 0.15 or more. Because all five region labels are permuted for every measure, the smallest attainable p-value is 1/120. Without Manavgat, most correlations are based on 2 to 12 directions; some rise, for example the supported-feature agreement to $\rho = 0.70$ (10 directions) and climatic distance to $\rho = 0.72$ (6 directions), but none of these samples can support inference
Without Manavgat (R8d)	Does one region drive the transfer results?	Without the eight directions that involve Manavgat, mean transfer was 0.566 on the original study areas (two of twelve directions below chance) and 0.631 on the 10 km collar (none below chance). The thermal gain was +0.015 and +0.032. Transfer is better without Manavgat, but it stays far below within-region skill
Equivalence on the original study areas (R8d)	Is the as-drawn thermal gain within ±0.05?	Yes, under all six resampling units that can be computed; the 90 % upper bounds range from 0.021 to 0.037. It is not within ±0.02 under any unit
Intervals for Table S27 (R8e)	Are the stratified LST associations supported?	On the 10 km collar, the Manavgat LST association within elevation deciles is 0.403 [0.343, 0.464], and after removing the linear effect of elevation 0.409 [0.350, 0.475]; both intervals exclude 0.5. On the original study area the values are 0.455 [0.396, 0.509] and 0.446 [0.387, 0.502], which include 0.5

Analysis	Question	Result
Capture of burned cells (R8b, R8g)	How many burned cells fall in the highest-scored cells?	In transfer, the 10 % of target cells with the highest thermal scores contained on average 10.3 % of the burned cells (range 0.7 % to 27.1 %), which is what a random ranking gives; eight of twenty directions were below it. The top 20 % contained 19.9 %. Within regions (5 km blocking) the top 10 % contained 35.5 % of the burned cells on average (18.9 % to 44.5 %), and the top 20 % contained 55.7 %
Pre-fire land cover (R8i, R8j)	Does the post-fire WorldCover v200 map (2021 images) affect the population or the results?	The 2020 map (v100) was fetched on each region's 30 m grid and aggregated to the same cells; the same procedure applied to v200 reproduces the pipeline's raster and every cell value of the modelling datasets exactly. With the 2020 map, at most 43 burned cells left or entered the natural-vegetation population in Manavgat, Bejís, Muğla and Evia. In Montiferru, 142 burned cells entered it: the share of burned modelled cells with a natural-vegetation fraction of at least 0.5 rose from 0.773 to 0.977, so the 2021 map left out about a fifth of the burned natural vegetation there. This share is computed on the modelled cells and is not the gate fraction of Table S15 (0.723 for Montiferru with the 2021 map), which the gate computes on its own inputs. The within-region thermal gain (5 km blocking) changed from +0.062, +0.045, +0.079, +0.148 and +0.099 to +0.060, +0.071, +0.083, +0.155 and +0.095 (Manavgat, Bejís, Muğla, Evia, Montiferru). Mean transfer changed from 0.527 to 0.513, and the mean thermal gain in transfer from +0.007 to +0.013; seven directions were below chance with both maps, and no direction changed side. Part of the class changes reflects the different algorithms of v100 and v200, not the fires
Second burned-area product (R8k, R8m)	Do the results depend on MCD64A1?	The MCD64A1 labels fetched again from Earth Engine reproduce the modelling datasets exactly in all five regions. VIIRS VNP64A1 v002 labels, built the same way, agree with them with a Jaccard index of 0.941, 0.950, 0.925, 0.978 and 0.865 (Manavgat, Bejís, Muğla, Evia, Montiferru). With VNP64A1 labels, the within-region thermal gain (5 km blocking) is +0.057, +0.055, +0.089, +0.147 and +0.082; the elevation association of Manavgat is 0.228 (0.232 with MCD64A1); mean transfer is 0.525 (0.527), the mean thermal gain in transfer +0.005 (+0.007), seven directions are below chance with both products, and no direction changes side
Terrain aspect (R8k, R8m, R8n)	Does aspect explain the thermal gain or the LST sign?	Northness and eastness from the Copernicus DEM were added to both feature sets. The within-region thermal gain (5 km blocking) became +0.051 [+0.030, +0.070], +0.044 [+0.025, +0.065], +0.073 [+0.047, +0.099] and +0.139 [+0.109, +0.173] in Manavgat, Bejís, Muğla and Evia, and +0.020 [−0.025, +0.066] in Montiferru, against +0.099 [+0.017, +0.186] without aspect there. Within northness deciles, the LST association is 0.652, 0.482, 0.317, 0.376 and 0.417 (raw 0.665, 0.478, 0.324, 0.377 and 0.370, with missing values filled by the median, so they differ from Table S9 in the third decimal), so aspect does not change its sign

Analysis	Question	Result
Weather reference (R8o)	Do the thermal channels follow weather anomalies?	Descriptively, across the five regions the mean Landsat LST anomaly follows the ERA5-Land anomaly of air temperature in the predictor window (Spearman $\rho = 0.8$) but not that of relative humidity ($\rho = 0.0$), and the mean TVDI difference does not follow humidity either ($\rho = 0.0$). With five regions this is descriptive only; it fits the reading that the channels describe temperature and surface state more than fuel moisture
Missing thermal values (R8p)	How many modelled cells lack a thermal value, and is this related to burning?	In the natural-vegetation population, 1.7 %, 6.5 %, 3.2 %, 4.5 % and 0.1 % of cells (Manavgat, Bejís, Muğla, Evia, Montiferru) have at least one missing thermal channel. The ROC-AUC of this missingness flag against burning is 0.493 to 0.512. Among all valid cells the share is larger (up to 57.5 % in Evia, mostly sea and coast)
Crossed random effects (R8l)	Does a model with source and target random intercepts change the thermal gain in transfer?	No. The estimated source and target variances are close to zero, so the fit is singular and reduces to a pooled interval with four degrees of freedom; it is a more conservative interval, not an independent confirmation. The mean gain is +0.007 [−0.035, +0.050] on the original study areas and +0.024 [−0.012, +0.060] on the 10 km collar (REML, t interval with four degrees of freedom); both include zero
Residual spatial correlation (R8l)	Do 5 km blocks remove the spatial dependence?	Not fully. The correlation of out-of-fold residuals between cell pairs is 0.58 to 0.83 below 1 km, 0.15 to 0.48 at 5 to 10 km (except Montiferru, −0.01), and falls below 0.05 at 10 to 20 km in Manavgat, Bejís and Muğla, at 20 to 40 km in Evia and at 5 to 10 km in Montiferru. Intervals at 5 km blocking may therefore be too narrow, especially in North Evia
Pooled against fold-averaged AUC (R8l)	Does pooling out-of-fold predictions inflate the within-region AUC?	No. The mean of the five fold AUCs is 0.004 to 0.018 higher than the pooled value in every region (thermal model, 5 km blocking)
Estimator intervals (R8l)	How uncertain are the four estimators of Table S25?	Pair-cluster intervals of mean transfer: 0.527 [0.461, 0.588], 0.527 [0.460, 0.594], 0.522 [0.456, 0.584] and 0.481 [0.419, 0.542]; of the thermal gain in transfer: +0.007 [−0.022, +0.034], −0.011 [−0.040, +0.016], −0.019 [−0.045, +0.007] and −0.021 [−0.057, +0.010]. All gain intervals include zero
Bejís pre-label cells (R8b)	Do the 48 cells that burned in the predictor window affect the results?	Ten of them are labelled burned. Without them, the within-region thermal gain was +0.050 at 1 km blocking (+0.056 with them) and +0.061 at 5 km (+0.045). Transfer in the eight directions that involve Bejís changed by at most 0.015, and their mean thermal gain from +0.011 to +0.007

A leave-one-block-out reference with a 10 km buffer was also tried. Its pooled AUC combines the predictions of different models, and because the burned cells lie in one or two blocks, the pooled value was dominated by calibration differences between the models. It is therefore not reported; the distance curve of Section S1.18, which scores a single model, answers the same question.

S2 Supporting tables

These tables give the values per region and per direction that the Results quote, so that every statement can be checked against the values behind it.

Table S9. Signed univariate AUC of each predictor against `burned`, by region. Natural-vegetation population; 10-cell (~5 km) spatial-block bootstrap, 1000 replicates, seed 42. The AUC is not folded to max(AUC, 1 − AUC). A value below 0.5 means that lower values rank burned cells higher, so it shows the direction of the association. Bold marks a region whose own interval excludes 0.5. Source `paper/labelfix_rerun/round5/tables/step9g_multi_aoi_feature_stability.csv` (the pipeline's Step9G five-region synthesis, sha256 c864cd7d…); checked row by row by `paper/code/appendix_tables.py`.

Feature	Manavgat	Bejís	Muğla	Evia	Montiferru
`elevation_mean`	**0.232** [0.179, 0.288]	**0.643** [0.558, 0.729]	**0.611** [0.532, 0.690]	0.541 [0.448, 0.626]	0.584 [0.395, 0.762]
`slope_mean`	**0.400** [0.340, 0.459]	0.521 [0.439, 0.605]	**0.637** [0.582, 0.686]	0.487 [0.418, 0.554]	**0.652** [0.506, 0.771]
`ndvi_mean`	0.564 [0.499, 0.628]	0.559 [0.497, 0.619]	**0.662** [0.616, 0.704]	**0.639** [0.575, 0.701]	0.586 [0.450, 0.704]
`lst_anomaly_mean`	0.509 [0.460, 0.560]	**0.418** [0.364, 0.480]	0.485 [0.395, 0.566]	**0.640** [0.567, 0.710]	0.395 [0.285, 0.535]
`current_lst_mean`	**0.665** [0.608, 0.719]	0.477 [0.401, 0.547]	**0.325** [0.271, 0.382]	**0.377** [0.301, 0.456]	0.370 [0.248, 0.513]
`current_tvdi_mean`	**0.677** [0.622, 0.733]	0.517 [0.429, 0.595]	**0.336** [0.275, 0.398]	**0.362** [0.285, 0.442]	**0.356** [0.233, 0.499]
`tvdi_difference_mean`	0.460 [0.409, 0.510]	0.512 [0.443, 0.583]	0.490 [0.396, 0.575]	0.519 [0.444, 0.589]	**0.378** [0.282, 0.497]
`downscaled_lst_mean`	**0.683** [0.626, 0.739]	0.484 [0.400, 0.560]	**0.307** [0.253, 0.366]	**0.377** [0.297, 0.459]	0.365 [0.240, 0.511]
`fused_lst_mean`	**0.666** [0.610, 0.721]	0.481 [0.404, 0.551]	**0.325** [0.272, 0.383]	**0.376** [0.300, 0.456]	0.370 [0.248, 0.513]

Definition of a reversal. A pair of regions is called a reversal only when their signed associations lie on opposite sides of 0.5 and the interval of each region excludes 0.5. This is stricter than requiring disjoint intervals. For example, for `current_lst_mean` in Manavgat and Bejís the two intervals are disjoint ([0.608, 0.719] against [0.401, 0.547]), but the Bejís

interval includes 0.5. Bejís therefore has no established direction, and this pair is a point reversal only.

Table S10. Cross-region reversals that meet the strict criterion. The criterion is applied to the intervals of Table S9. Difference intervals are from the paired 10-cell block bootstrap of `paper/code/ems_inference_multiplicity.py` (`paper/labelfix_rerun/inference/reversal_family_holm.csv`, 1000 replicates), which flags the same fourteen pairs. Checked row by row by `paper/code/appendix_tables.py`.

Feature	Region A	AUC	Region B	AUC	Difference [95 % CI]
`elevation_mean`	Manavgat	0.232	Bejís	0.643	+0.411 [+0.312, +0.509]
`elevation_mean`	Manavgat	0.232	Muğla	0.611	+0.379 [+0.274, +0.476]
`slope_mean`	Manavgat	0.400	Muğla	0.637	+0.237 [+0.158, +0.313]
`slope_mean`	Manavgat	0.400	Montiferru	0.652	+0.252 [+0.100, +0.389]
`lst_anomaly_mean`	Bejís	0.418	Evia	0.640	+0.222 [+0.129, +0.316]
`current_lst_mean`	Muğla	0.325	Manavgat	0.665	+0.340 [+0.260, +0.413]
`current_lst_mean`	Evia	0.377	Manavgat	0.665	+0.288 [+0.195, +0.384]
`current_tvdi_mean`	Muğla	0.336	Manavgat	0.677	+0.341 [+0.253, +0.422]
`current_tvdi_mean`	Evia	0.362	Manavgat	0.677	+0.315 [+0.221, +0.413]
`current_tvdi_mean`	Montiferru	0.356	Manavgat	0.677	+0.322 [+0.161, +0.453]
`downscaled_lst_mean`	Muğla	0.307	Manavgat	0.683	+0.376 [+0.295, +0.451]
`downscaled_lst_mean`	Evia	0.377	Manavgat	0.683	+0.307 [+0.209, +0.405]
`fused_lst_mean`	Muğla	0.325	Manavgat	0.666	+0.341 [+0.261, +0.413]
`fused_lst_mean`	Evia	0.376	Manavgat	0.666	+0.291 [+0.198, +0.386]

There are fourteen reversals between pairs, across seven features, and thirteen of them involve Manavgat. Under the original label there were three, across two features: elevation and the LST anomaly. The feature-removal analysis of Section 2.11 keeps these two and does not re-select. Twenty-six further pairs reverse at the point estimate only, and they are not counted. The strict criterion reduces the number of findings: a difference interval on the pair (the test of Section S1.12) would support seventeen further reversals. None of the fourteen is corrected for multiplicity; Table S30 gives the corrected counts.

S2.1 The transfer matrix in precision-recall terms

ROC-AUC is used in the main text so that the results can be compared with the susceptibility literature. A susceptibility map is used to rank areas, so precision-recall measures its practical value. At target prevalences of 7.0 to 28.7 %, the two measures can differ strongly. Values are read from the step9b exports of the re-frozen outputs (`paper/labelfix_rerun/round5/tables/corrected/*/step9b_metrics.json`).

Table S11. Thermal transfer, PR-AUC against the no-skill baseline. The baseline is the burned prevalence of the target. Lift is PR-AUC divided by this baseline; a lift of 1 is no better than a random ranking. Ordered by lift. Checked row by row by `paper/code/appendix_tables.py`. PR-AUC intervals at 2-cell and 10-cell blocking for every direction are in `paper/labelfix_rerun/round7/r7a_pr_auc_10cell.csv`.

Direction	ROC-AUC	PR-AUC	No-skill	Lift
Evia → Manavgat	0.677	0.321	0.143	**2.24**
Manavgat → Evia	0.654	0.407	0.287	1.42
Bejís → Montiferru	0.594	0.289	0.212	1.36
Evia → Montiferru	0.647	0.283	0.212	1.34
Bejís → Muğla	0.618	0.093	0.070	1.33
Muğla → Evia	0.653	0.379	0.287	1.32
Montiferru → Bejís	0.548	0.093	0.072	1.28
Montiferru → Muğla	0.619	0.089	0.070	1.27
Evia → Muğla	0.577	0.086	0.070	1.23
Muğla → Bejís	0.583	0.088	0.072	1.22
Manavgat → Montiferru	0.518	0.243	0.212	1.15
Montiferru → Evia	0.586	0.317	0.287	1.11
Muğla → Montiferru	0.531	0.214	0.212	1.01
Manavgat → Muğla	0.438	0.060	0.070	**0.85**
Evia → Bejís	0.448	0.059	0.072	**0.82**
Montiferru → Manavgat	0.404	0.113	0.143	**0.79**
Bejís → Evia	0.383	0.226	0.287	**0.79**
Manavgat → Bejís	0.396	0.054	0.072	**0.75**
Muğla → Manavgat	0.345	0.101	0.143	**0.70**
Bejís → Manavgat	0.314	0.098	0.143	**0.68**
Mean	**0.527**	**0.181**	**0.157**	**1.13**

Seven directions are below their own no-skill baseline, and only one is above twice its baseline. The seven are the same seven that are below chance on ROC-AUC, as expected for

a reversed ranking. At 1 km blocking, all seven have PR-AUC intervals fully below the baseline. At 5 km blocking two do: Bejís to Manavgat [0.067, 0.132] and Muğla to Manavgat [0.071, 0.135], against 0.143. Section 3.4 shows that the number of directions below chance depends largely on the study areas.

S2.2 The same-geography event pair, in full

Section S1.13 reports this comparison, shows that its elevation reversal is caused by the study area and explains why its thermal channels are not evaluated. The values per feature are kept here because this comparison led to the frame test.

Table S12. Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against `burned`, not folded to max(AUC, 1 − AUC); 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section S1.13). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). Blocks of 5 km with burned cells: 70 in 2021 and 11 in 2022. The note to Table 2 sets sixteen as the minimum for this blocking, so the 2022 intervals are indicative, like the 20-cell row of Table 2. The 2022 fire is also one compact scar, so its eleven blocks are next to each other. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	**0.611 [0.532, 0.690]**	**0.296 [0.230, 0.355]**	**bootstrap-supported**
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

S2.3 The evaluation frames, and the signed associations they produce

Section 3.4 gives these results, and Fig. 1b of the main text maps the five study areas with their burned cells, scar frames and 10 km collars. In Muğla one burned area reaches the eastern edge of the study area (15 burned cells on the edge), so its collars are cut there. The values per region are given here.

Table S13. Geometry of the five study areas. Natural-vegetation population. Distance is the Euclidean distance to the nearest burned cell on the 500 m grid, at 0.45 km per cell. Corrected Manavgat label. Computed by `paper/code/appendix_tables.py` from the step8a tables (read through `paper/code/_canonical.py`, which checks the sha256 of each file), as in `paper/code/verify_aoi_frame.py`.

Region	cells	burned	median distance to burned	share beyond 10 km
Manavgat	20,511	2,935	13.1 km	**58.5 %**
Bejís	15,190	1,100	13.5 km	**63.1 %**
Muğla	41,730	2,911	11.3 km	55.3 %
Evia	9,298	2,664	8.0 km	43.7 %
Montiferru	2,544	539	2.7 km	**2.1 %**

In Manavgat, the median elevation of the modelled cells is 330 m within 5 km of the fire (burned cells included), 995 m at 10 to 20 km and 1,273 m at 20 to 50 km; the burned cells have a median of 287 m (printed by `paper/code/verify_aoi_frame.py`, stage 1b).

Table S14. Signed univariate AUC, original study area against a 10 km collar. Point estimates. The intervals used for the reversal test are given in the text and in `collar_frame_bootstrap.csv` (10-cell blocks, 1000 replicates, seed 42). Signed and not folded to max(AUC, 1 − AUC), so a value below 0.5 shows the direction of the association. The collar removes no burned cells in any region. Full frame from the source of Table S9; collar from `paper/labelfix_rerun/code/aoi_frame_auc_frozen_mugla.csv` (the collar analysis with Muğla read from its canonical file). Checked row by row by

paper/code/appendix_tables.py. On every row, Manavgat is on the other side of 0.5 from the other regions, also on the collar.

Signed univariate AUC	Manavgat	Bejís	Muğla	Evia	Montiferru	straddles 0.5
elevation, full frame (Table S9)	**0.232**	0.643	0.611	0.541	0.584	**yes**
elevation, 10 km collar	**0.376**	0.614	0.606	0.648	0.581	**yes**
current LST, full frame	**0.665**	0.477	0.325	0.377	0.370	**yes**
current LST, 10 km collar	**0.522**	0.405	0.332	0.286	0.376	**yes**
current TVDI, full frame	**0.677**	0.517	0.336	0.362	0.356	**yes**
current TVDI, 10 km collar	**0.527**	0.454	0.342	0.250	0.361	**yes**

Table S15. Region summary: populations and gate results. Counts from the Step 8A dataset statistics of each region; gate fractions from the burned-landcover gate output of each region. TSG = the natural-vegetation population (tree, shrub and grass). The TSG columns use the modelled population, burnable_tree_shrub_grass and valid_for_modeling == True, on which every model in this paper was fitted and scored. Counts are computed from the step8a tables, and gate fractions are read from the gate output of each region (paper/labelfix_rerun/round5/tables/corrected/gates/). Checked row by row by paper/code/appendix_tables.py.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural-veg fraction	Gate verdict
Manavgat 2021	24,150	24,087	3,046	0.126	20,511	2,935	0.143	0.955	pass
Bejís 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (extended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Montiferru 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

Table S16. Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$). Generated by `paper/code/table_b9.py` from the re-frozen outputs (`paper/labelfix_rerun/round5/matrix20_official.csv`).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.396 [0.373, 0.422]	0.450 [0.423, 0.477]	0.467 [0.441, 0.491]
Bejís→Manavgat	0.314 [0.296, 0.332]	0.302 [0.282, 0.323]	0.406 [0.388, 0.423]
Manavgat→Muğla	0.438 [0.418, 0.456]	0.427 [0.408, 0.444]	0.417 [0.398, 0.436]
Muğla→Manavgat	0.345 [0.331, 0.359]	0.485 [0.468, 0.502]	0.476 [0.460, 0.493]
Manavgat→Evia	0.654 [0.633, 0.676]	0.529 [0.504, 0.553]	0.504 [0.481, 0.528]
Evia→Manavgat	0.677 [0.658, 0.697]	0.404 [0.386, 0.421]	0.417 [0.399, 0.435]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.404 [0.385, 0.422]	0.388 [0.367, 0.408]	0.436 [0.416, 0.456]
Manavgat→Montiferru	0.518 [0.474, 0.563]	0.527 [0.484, 0.568]	0.505 [0.461, 0.547]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

S2.4 The transferability-diagnostics tables

Table S17. All similarity measures against raw thermal transfer (20 ordered directions). Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with n = 12 exist only for the four-region subset, because these measures were not produced for Montiferru. The supported conditional rows use the 18 directions with at least one supported feature. Every row is from `paper/labelfix_rerun/round5/s6_diagnostics_20.csv`, checked by `paper/code/appendix_tables.py`. The twenty measures were defined on 8 August 2026 and were not re-selected after the label correction.

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	P(y|x) conditional	+	18	+0.52 [−0.27, +0.87]	no
Cosine, supported features	P(y|x) conditional	+	18	+0.49 [−0.24, +0.87]	no
Cosine, all 9 features	P(y|x) conditional	+	20	+0.44 [−0.29, +0.80]	no
Vector Spearman, all 9	P(y|x) conditional	+	20	+0.43 [−0.21, +0.80]	no
Agreement count, all 9	P(y|x) conditional	+	20	+0.47 [−0.29, +0.80]	no
Schoener's D, 1-D mean	P(x|y=1) niche	+	20	+0.10 [−0.53, +0.63]	no
Warren's I, 1-D mean	P(x|y=1) niche	+	20	+0.19 [−0.43, +0.77]	no
Schoener's D, PCA-2D	P(x|y=1) niche	+	20	+0.03 [−0.59, +0.63]	no
Warren's I, PCA-2D	P(x|y=1) niche	+	20	−0.03 [−0.66, +0.56]	no
Mahalanobis, burned centroids	P(x|y=1) niche	−	20	−0.22 [−0.82, +0.42]	no
Domain-classifier AUC	P(ix) marginal	−	20	−0.33 [−0.81, +0.36]	no
Predictor-space mean dissimilarity	P(ix) marginal	−	12	−0.13 [−0.67, +0.37]	no
Predictor-space p95 dissimilarity	P(ix) marginal	−	12	−0.15 [−0.68, +0.24]	no
Fraction inside weighted AoA	P(ix) marginal	+	12	+0.21 [−0.48, +0.62]	no
Fraction inside unweighted support	P(ix) marginal	+	12	+0.08 [−0.92, +0.76]	no
Climatic distance	P(ix) marginal	−	12	+0.03 [−0.84, +0.84]	no
Geographic distance	geographic	−	12	−0.17 [−0.87, +0.84]	no
Regime distance, log effective-N	P(y) structure	−	20	+0.11 [−0.67, +0.80]	no
Regime distance, largest share	P(y) structure	−	20	+0.11 [−0.66, +0.77]	no
Vector Spearman, supported (≥3 feats) ‡	P(y|x) conditional	+	6	+0.96 [+0.85, +0.96]	degenerate interval; not interpreted

Table note. A null row means that the measure was not shown to order transfer in this design. It does not mean that the measure cannot order transfer. With ten independent region pairs the power is low, and the intervals are wide enough to include moderate true correlations in either direction. ‡ The last row belongs to the set of measures fixed in advance, so it stays in the table. It is defined on six directions only, and its interval is degenerate (the upper bound equals the point estimate), so it is not interpreted and cannot be compared with the other nineteen.

Equal-sample check. The families use different samples: twelve directions for the marginal, applicability, climatic and geographic rows, eighteen for the supported conditional rows, and twenty for the others. Every row was recomputed on the common subset of twelve directions,

and again on the sixteen directions where the supported conditional rows are defined. Every interpretable row still includes zero. On the common twelve, the supported conditional rows give +0.41 [−0.42, +0.88] and +0.46 [−0.56, +0.84]. The published values are reproduced to 4.4×10^{-5}. Source: `paper/labelfix_rerun/round5/out_official/diagnostics_common_subset.json`.

Table S18. The most and least environmentally similar pairs, on both frames. Schoener's *D* is computed over burned cells only, so it does not change with the collar. Transfer values are the two ordered directions of each pair. Ranks are out of the twenty directions on the collar. Corrected Manavgat label. Schoener's *D*, per-feature *D* and transfer on the original study areas are read from the source of Fig. 8, `paper/labelfix_rerun/round5/out_official/figure_contrast_pairs.json`. Collar transfer and ranks are from the source of Table S8, `paper/labelfix_rerun/round5/collar/aoi_frame_transfer.csv`, and the AoA shares from `paper/labelfix_rerun/round5/collar/aoa_directed_pair_summary.csv`. Every cell is checked against these files at build time (`paper/figures/fig8_contrast_pairs.py`).

	Manavgat and Muğla	Bejís and Montiferru
Schoener's *D*, mean 1-D	**0.799** (highest)	**0.479** (lowest)
per-feature *D*	0.74 to 0.87	0.23 to 0.77
transfer, frames as drawn	0.438, 0.345	0.594, 0.548
transfer, 10 km collar	0.493, 0.433	0.669, 0.624
rank of 20 on the collar, from the bottom	4th, 2nd	16th, 12th
target cells inside the AoA	0.876, 0.531	n/a

S3 Protocol detail and limitations

These protocols are given here in full. They are needed only to check the related results; the main argument can be followed without them. Sections 2.2 to 2.4, 2.7, 2.9, 2.10 and 2.13 summarise them.

S3.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate

Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`), retrieved through Google Earth Engine (Gorelick et al. 2017). Its omission and commission errors (Boschetti et al. 2019) limit every model here. The product is exported onto the 30 m EPSG:4326 reference grid used by the rest of the pipeline. Each native ~500 m observation is therefore copied to the 30 m pixels below it.

The analysis grid is rebuilt from these pixels. The 30 m grid is divided into non-overlapping blocks of `round(500 / 30) = 17 × 17` pixels, which gives a nominal cell edge of 510 m. Each cell is identified by its block indices `row_500m` and `col_500m`. This approximates the MODIS sinusoidal cell but does not reproduce it, because it is anchored to the Landsat/EPSG:4326 reference grid. Blocks at the edges are cut. The cell is square in degrees but not on the ground: about 510 m from north to south and 390 to 407 m from east to west. Its area is therefore 0.199 to 0.208 km², against 0.215 km² for the native MODIS cell (463 m edge). Every block size in this paper refers to the north-south dimension. The collar step of 0.45 km per cell (Section 2.12) lies between the two cell dimensions, so collar radii are nominal to within about 15 %.

The burn date of a cell is the mode of the burned sub-pixel day-of-year values in the block, and it is tested against the label window of the region. The exported raster has no burned pixels outside the window, so in this dataset one burned sub-pixel inside the window makes the cell burned. This can extend each scar by up to one cell at its edge, and it can mark a water cell as burned where the product does. The primary population keeps only cells dominated by natural vegetation, so water is excluded. The fraction of burned sub-pixels that agree with the modal date is recorded as `burn_date_pixel_agreement_fraction`, but no threshold is

applied. The label never affects whether a cell is modelled: unburned, all-no-data and out-of-window cells all stay in the negative class.

Earlier burning is handled differently in each region; this is given in Section S3.1.1. The only historical exclusion applied by design removes the 2021 Muğla scar from the 2022 experiment of Section S1.13.

Before any modelling, each region passes a gate that asks one question: what fraction of the burned cells is dominated by natural vegetation? A region is admitted as a wildfire case when this fraction is at least 0.50 and at least 30 burned cells are present. It is rejected as a cropland control when the cropland fraction is at least 0.50. The gate uses ESA WorldCover classes aggregated to the same cells. It separates burning of natural fuel from stubble burning after harvest, which MCD64A1 does not distinguish. The results are given in Section 4.1.

S3.1.1 Pre-label and historical burning, by region

Two kinds of earlier burning can affect a label. *Pre-label burning* is burning inside the predictor window of the region, which the pipeline can remove. *Historical burning* is burning in the five previous years, which no region screens for. Table S34 gives both, recomputed from the MCD64A1 export of each region (`paper/code/ems_labels_1_prelabel.py`, `ems_labels_2_history.py`; outputs `paper/labelfix_rerun/labels/r1_prelabel.json`, `r2_history.json`).

Table S34. Earlier burning by region, natural-vegetation population.

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	… of which burned again in the event
Manavgat 2021	none needed (no pre-label burning)	0	0	0
Bejís 2022	exclusion not applied	48 (49 in all valid cells; 10 burned again)	163	0

Region	Pre-label cells removed	Pre-label burned cells retained	Historical-burn cells (five years)	… of which burned again in the event
Muğla 2021	49	0	194	25
North Evia 2021	16	0	366	88
Montiferru 2021	61	0	58	32

Excluding the historically burned cells lowers the mean paired transfer contribution from +0.007 to +0.003 (at 10-cell blocking: 5 positive, 4 negative and 11 uncertain verdicts, against 6, 5 and 9). The within-region gain stays positive with interval support in every region: +0.062 (no change), +0.055, +0.088, +0.162 and +0.111 for Manavgat, Bejís, Muğla, Evia and Montiferru.

S3.1.2 The Manavgat label correction

The Manavgat 2021 label is corrected. The label first used for Manavgat was exported on 8 July 2026, before the month-alignment fix of the MCD64A1 query in the pipeline (commit 183be42, 11 July), and the export was not renewed, so it missed the first four days of the fire, 28 to 31 July. The error came from this old export; the code itself was correct. The corrected label only adds burned cells: no cell that burned under the original label becomes unburned, and no predictor or validity flag changes. In the primary population, the number of burned cells rises from 784 to 2,935. The Manavgat outputs were re-frozen with the pipeline at commit 6381f4c. Two small changes were made: a one-line patch to the window-closure diagnostic, and a runtime setting of the Earth Engine project (`thermaltwin`). Python 3.12.10 and scikit-learn 1.9.0 were used (versions in `ENVIRONMENT.md`). The correction, the re-freeze and the control run were carried out by the corresponding author with AI-assisted code (see the statement on the use of generative AI in the main article). They were not repeated by the author of the pipeline. This pipeline version writes one extra column, a flag for burning in earlier years, and it excludes no Manavgat cell. In a control run, the original label was re-frozen with the same code and environment. It reproduced the published outputs to within

10^{-5}. The exceptions are listed in the manifest: file paths, the column list, and one older pair that differs at the level of random-forest thread nondeterminism. Differences between the two runs are therefore caused by the label alone. The reproduction check of Section 2.13 covers the re-frozen outputs. The manifest, hashes and scripts are released in `paper/data/manavgat_2021/refreeze/`.

S3.1.3 Effect on Table 2

The Manavgat rows use the corrected label. It adds burned cells (784 to 2,935 in the primary population) and changes no predictor (Section 2.2). The rows become stronger. Absolute AUCs rise by 0.04 to 0.11 across the three block sizes, and the gain holds at every scale. The 1 km interval becomes narrower, from [+0.055, +0.079] to [+0.060, +0.073], because the region now has 814 blocks of 2 cells with burned cells instead of 235.

S3.2 Transferability diagnostics versus transfer

Twenty similarity measures from five families are each rank-correlated with the same quantity, the raw thermal transfer AUC over the twenty ordered directions, with one common pair-based bootstrap. The families are marginal predictor-distribution measures P(x), burned-niche overlap P(x|y=1), burned-pattern structure P(y), and conditional feature-response direction P(y|x).

The *marginal* family includes area-of-applicability dissimilarity in scaled, importance-weighted predictor space (Meyer and Pebesma 2021, 2022; Ludwig et al. 2023), climatic and geographic distance, and a domain classifier trained to separate source cells from target cells. The *niche-overlap* family includes Schoener's D (Schoener 1968) and Warren's I (Warren et al. 2008) over the burned cells of each region. The *regime* family includes counts of burned patches and effective patch counts. The *conditional* family is the sign-agreement index: the

fraction of predictors whose signed association points the same way in both regions, computed over all predictors and over the predictors with interval support.

Two properties of this design limit what it can show. First, the set of measures was recorded on 8 August 2026, together with its results under the original label (commit abd6a4b on the `history-archive` branch of the public repository, which keeps the development history without internal notes). It was not re-selected after the label correction. This is a record and was not pre-registered. Second, with five regions the effective sample is ten unordered pairs, and nineteen measures are evaluated at this power without family-wise error control. Where a measure passes an interval test but not a Bonferroni threshold, both results are reported.

S3.3 Same-geography event-to-event comparison (Muğla 2021 versus 2022)

Muğla is the only region in which a second fire is analysed on the same study area and grid. This allows the direction of feature-label associations to be compared with the place held fixed. The 2022 windows are set relative to the 2022 fire: a 58-day predictor window that closes the day before ignition, and a 49-day label window that opens on that day. These lengths are the same as in 2021.

This is not a clean design for transfer over time, and it is not presented as one. The 2022 fire started about six weeks earlier in the season than the 2021 fire, so year and seasonal phase are confounded, and no difference can be attributed to the year alone. The comparison is between two fires in one place. The pair is kept out of the twenty-direction matrix, and this is enforced in the released code.

The 2022 population is the 2021 population without the 2021 scar; Section S1.13 describes what follows from this.

Table S19. Study regions, study areas and time windows. Bounding boxes are in EPSG:4326, as recorded. Window lengths in brackets are day counts, both ends included. The baseline years are the four years before each predictor window, with the same calendar dates.

Region	Bounding box (lon min, lat min, lon max, lat max)	Predictor window	Label window	Baseline years
Manavgat 2021 (Türkiye)	31.05, 36.72, 31.85, 37.35	2021-06-01 to 2021-07-27 (57 d)	2021-07-28 to 2021-08-31 (35 d)	2017, 2018, 2019, 2020
Bejís 2022 (Spain)	-1.05, 39.68, -0.35, 40.15	2022-06-15 to 2022-08-14 (61 d)	2022-08-15 to 2022-09-30 (47 d)	2018, 2019, 2020, 2021
Muğla 2021 (Türkiye)	27.1, 36.6, 28.9, 37.45	2021-06-01 to 2021-07-28 (58 d)	2021-07-29 to 2021-09-15 (49 d)	2017, 2018, 2019, 2020
North Evia 2021 (Greece)	23.05, 38.55, 23.85, 39.15	2021-06-05 to 2021-08-02 (59 d)	2021-08-03 to 2021-09-30 (59 d)	2017, 2018, 2019, 2020
Montiferru 2021 (Italy)	8.45, 40.05, 8.75, 40.27	2021-05-25 to 2021-07-23 (60 d)	2021-07-24 to 2021-08-31 (39 d)	2017, 2018, 2019, 2020

S3.4 Predictor provenance, compositing and the downscaler

Ten predictors are used, four baseline and six thermal. All are summarised per cell over the predictor window, through the chain shown in Fig. 2. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (`LANDSAT/LC08/C02/T1_L2`). They are quality-screened per pixel with the `QA_PIXEL` band; the water bit is kept on purpose. The coarse thermal input is `MODIS/061/MOD11A1`.

- NDVI is the median of Landsat surface reflectance over the predictor window. It is the only baseline predictor that changes over time.
- Elevation and slope come from the Copernicus DEM GLO-30. Its heights are referenced to the EGM2008 geoid (EPSG:3855) in metres, according to the product handbook. The pipeline applies no datum conversion.
- Land cover is ESA WorldCover v200 (Zanaga et al. 2022), used as the dominant class code.
- Current LST is the median of Landsat surface temperature over the predictor window, in °C.

- LST anomaly is a z-score of the current LST median against the four baseline years. It is set to no-data where the baseline standard deviation is below 1.0 °C or there are too few observations.
- TVDI and TVDI difference. The Temperature-Vegetation Dryness Index (Sandholt et al. 2002) is computed in the LST-NDVI space. The wet and dry edges are the 2nd and 98th LST percentiles within each of twenty NDVI bins, and the index is limited to [0, 1]. The edges are percentiles of the LST values in a given scene, so they are fitted for each study area and window. A TVDI of 0.5 therefore means a different moisture state in each region. TVDI difference is the raw anomaly of this index against the same four baseline years: `tvdi_difference = current_tvdi − mean(baseline_tvdi)`. It is in index units, while the LST anomaly is in standard deviations, so the two are reported together.
- Downscaled LST and fused LST. A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals the observed Landsat LST where that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler uses coordinates as inputs. This is the only way in which coordinate information enters a feature set from which Section 2.13 excludes coordinates. Section S1.7 gives the gain without these two channels.

The six thermal channels carry about two signals. In the primary population, current, fused and downscaled LST are correlated at 0.97 to 1.00 in every region. Current LST and TVDI are correlated at 0.88 to 0.98, and the two anomaly channels at 0.71 to 0.94. Two principal components of the six channels explain 92 % to 98 % of their variance, and their participation ratio is 1.6 to 1.8 (`paper/labelfix_rerun/labels/r4_dimensionality.json`).

Table S20. Input data. All retrieved through Google Earth Engine.

Input	Product and version	Native resolution	Use	Reference
Burned area	MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`)	500 m, monthly	label	(Giglio et al. 2018, 2021)
Surface reflectance and temperature	Landsat 8 Collection 2 Level-2 (`LANDSAT/LC08/C02/T1_L2`), `QA_PIXEL` screening, water bit preserved	30 m	NDVI, current LST, anomaly, TVDI	(U.S. Geological Survey Earth Resources Observation and Science (EROS) Center 2020)
Coarse LST	MODIS MOD11A1 v061 (`MODIS/061/MOD11A1`)	1 km, daily	downscaled and fused LST (gap-fill 0.11 % to 9.70 %)	(Wan et al. 2021)
Terrain	Copernicus DEM GLO-30 (EGM2008 heights)	30 m	elevation, slope	(European Space Agency 2022)
Land cover	ESA WorldCover v200 (2021)	10 m	dominant class, gate, population	(Zanaga et al. 2022)

Compositing uses the median over the predictor window. The anomaly channels refer to four baseline years (Table S19). WorldCover 2021 is later than four of the five fires (Section 2.4).

Table S36. Landsat 8 observations per window. Scenes = Landsat 8 Collection 2 Level-2 scenes whose footprint intersects the study area. Clear observations = number of clear observations per pixel after the pipeline's QA mask (`src/step3_landsat_lst.py`), median and 10th to 90th percentile over the study area at 300 m. The pipeline also filters the predictor window by calendar month, so its counts can be slightly lower. Landsat 9 is not used. Source `paper/labelfix_rerun/round8/r8k_landsat_observations.csv`.

Region	Predictor window scenes	Clear observations, predictor window	Clear observations, baseline years (medians)
Manavgat 2021	14	6 [3 to 10]	7, 7, 6, 6
Bejís 2022	16	3 [2 to 6]	4, 4, 3, 2
Muğla 2021	18	6 [3 to 10]	6, 6, 6, 6
North Evia 2021	8	4 [3 to 4]	4, 3, 2, 1
Montiferru 2021	4	3 [2 to 4]	4, 3, 2, 3

The anomaly channels therefore rest on few clear observations per year, one to seven depending on region and year, which adds noise to the four-year baseline.

S3.5 Limitations, in full

Section 4.6 lists the limitations that affect the conclusions. All fourteen are given here.

(i) No weather predictors are used, so the behaviour of a combined thermal and weather predictor set is not known.

(ii) The same-place comparison covers one region only, and there, year and seasonal phase are confounded. This cannot be resolved in this study area. The two fires are 42 days apart in median burn day-of-year, neither year has a second fire at the phase of the other, and a calendar-matched comparison would have nine burned cells, against a gate minimum of thirty. The 2022 population also has only eleven blocks with burned cells, against the minimum of sixteen (Section S1.13). Its 331 burned cells do not resolve the thermal reversals at interval level, and the pair keeps the place fixed but not the population. The two Muğla populations were computed with the unmodified pipeline code and the same fixed environment, not read from a frozen export.

(iii) All labels come from one burned-area product, MCD64A1 (Giglio et al. 2018). Its omission and commission errors (Boschetti et al. 2019) limit every model here. Its accuracy in the Mediterranean was assessed against reference perimeters (Katagis and Gitas 2022). A second 500 m product, VIIRS VNP64A1 (Giglio 2024), was used as a label check (Table S35), and no conclusion changed. Independent fire perimeters were not used.

(iv) Evia has the most unusual class balance, even on the extended study area. Its TSG prevalence is 0.287, the highest in the cohort, against 0.070 to 0.212 in the other four regions.

(v) Each region contributes one fire season, so regional concept shift is confounded with the weather of each fire. Labels from several years are needed to separate them.

(vi) Point estimates depend on the software version. Under scikit-learn 1.5.2 instead of 1.9.0, single-direction AUCs of the frame-transfer matrix move by up to 0.023 for the thermal model and 0.047 for the baseline, and paired differences by up to 0.045. The means move by at most 0.003. One direction changes interval support on each of the original and 10 km frames (`paper/labelfix_rerun/round7/sklearn152/compare_vs_1_9_0.json`). All reported numbers come from one verified version, and exact reproduction needs the archived environment.

(vii) The low transfer into Manavgat is located but not explained. Manavgat is the weakest target (0.435), has the largest matched shortfall (+0.319) and has the most unusual univariate profile in the cohort. Two processing causes were tested, and neither explains it. The quality screening of its coarse thermal input moves no signed association by more than 0.005 (Section S1.5), and its elevation reversal remains on the collar (Section 3.4). Holding elevation reverses its raw LST association (Section S1.11), while holding aspect does not. This rests on one region and one fire.

(viii) Measures based on interval support are less stable than the point estimates. Five to six of the pipeline's interval bounds lie within 0.01 of 0.5, under both labels. One such flag, the NDVI of Manavgat, moved the supported-feature cosine on the original study areas from $\rho = 0.49$ to 0.70, when two equally valid bootstrap streams disagreed on it. The transfer verdicts are more stable. Across five bootstrap seeds, every verdict at 1 km blocking is stable, and the paired split of twelve positive, seven negative and one uncertain holds on every seed. At 5 km blocking, one level verdict

and two paired-delta verdicts change with the seed. Exact counts of supported directions should be read at this precision.

(ix) The five study areas are not comparable, and this cohort cannot fully correct this (Section 4.4). The collar results are reported next to the original results and do not replace them, because the collar radius is also a choice, and 5 km and 10 km do not agree exactly (0.591 against 0.589). The study areas were fixed earlier in the pipeline, in `repo/`, so they can be restricted but not extended. Montiferru, whose rectangle is already close to the fire, cannot be given distant cells. Their independence from the outcomes can also not be fully documented. Only the Montiferru rectangle follows a rule, and the Manavgat rectangle is dated before its first gate result. For Bejís and Muğla, the version history cannot show that the rectangle was fixed before the first gate result (Section 3.1). Future studies should fix the study area by a stated accessible-area rule (Barve et al. 2011) before any predictor is computed.

(x) Only four estimators were compared. The main results use a random forest with unlimited depth, which fits local structure well and extrapolates poorly. Section S1.8 shows that the transfer result does not depend on this choice. Three other estimators, including a penalised linear model, range from 0.481 to 0.527 and have eleven to thirteen of twenty directions above chance. Their pair-cluster intervals overlap (Table S35), so the estimators are not ranked. Other estimator classes were not tried.

(xi) Burn timing and elevation are confounded in Manavgat. The cells that burned in the first four days lie at a median of 219 m, against 512 m for the later burns and 1,004 m for unburned cells. The effects of timing and elevation cannot be separated in these data.

(xii) The quality-screening comparison differs in code version as well as in screening. The current pipeline step7 refuses the unscreened MODIS input, because the raster has no nodata tag and 8.1 % exact zeros. The unscreened version therefore uses the step7 of export time, and the screened version the current one. The two agree on elevation to four decimals and on every other signed AUC to within 0.005; the largest change is −0.0044 for downscaled LST. The within-region gain moves from +0.067 to +0.068. The confound could hide an effect only if two effects cancelled (Section S1.5).

(xiii) The similarity tests rest on ten independent region pairs. Five regions give twenty ordered directions, but the two directions of a pair share both regions and are not independent. All results of Section S1.21, positive and negative, have this low power. With ten pairs, a moderate true ordering would usually be missed.

(xiv) The frame cost rests on few scars. The full comparison of Section 3.3 is defined on seven scars in three regions. Its scar-level intervals are too narrow, because row A is repeated for each scar of a region. The region-level estimates (+0.137 over three regions, +0.160 over five) should be quoted.

S3.6 Leakage control and reproducibility, in full

A fixed set of forbidden columns is checked at every model fit. Coordinates (`lon`, `lat`, `row`, `col` and their normalised forms), every burn-date and label-provenance column, and the agreement fraction are excluded from all feature sets. The check is an assertion in the code. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking keeps neighbouring cells in the same fold.

Reproducibility. All random processes use seed 42, and every bootstrap uses 1000 replicates. The reproduction check is described in Section S3.6.2.

S3.6.1 Resampling units

Uncertainty is estimated with a spatial-block bootstrap. Let $\beta_1, \ldots, \beta_M$ be the blocks of Eq. (1) of the main text that contain cells of F. Replicate b draws M indices u_{bm} uniformly with replacement:

$$F^{*b} = \biguplus_{m=1}^{M} \beta_{u_{bm}}, \qquad \mathrm{CI}_{95} = [Q_{0.025}\{\theta(F^{*b})\}_b,\ Q_{0.975}\{\theta(F^{*b})\}_b]. \qquad (S1)$$

Here θ is the statistic and Q gives the 2.5 and 97.5 percentiles over 1000 replicates, seed 42. A replicate with only one class is dropped. Differences are formed within each replicate, so they are paired. The block size of each interval is given with the result. Because blocks are resampled, the reliability of an interval depends on the number of blocks with at least one burned cell. These counts are reported with the intervals. Where a verdict rests on too few such blocks, it is reported as indicative.

The twenty transfer directions are not independent, because each region appears in eight of them. The *pair-cluster bootstrap* therefore resamples the ten unordered region pairs. Each pair carries its set π_p of ordered directions, and the carried values δ_{st} are averaged:

$$\bar{\delta}^{*b} = \frac{\sum_{m=1}^{10} \sum_{(s,t)\in\pi_{u_{bm}}} \delta_{st}}{\sum_{m=1}^{10} |\,\pi_{u_{bm}}|}. \qquad (S2)$$

The interval has the same percentile form, with 1000 replicates for the direction-level intervals of Sections 3.4 and 3.5, and 20,000 for the unit comparison of Table S31. Clustering by target region replaces π_p by the four directions that share a target. A quantity with one value per held-out scar or target region gets a Student t interval over these n units:

$$\bar{x} \pm t_{0.975,\,n-1}\, s_x/\sqrt{n}. \qquad (S3)$$

The effective sample is therefore ten pairs or five regions for direction-level intervals, and at most seven scars from three regions for scar-level intervals.

S3.6.2 Reproduction of the re-frozen outputs

The transfer analysis runs in an environment separate from the pipeline. The reproduction check of the repository therefore refitted every within-region model and all twenty CORAL transfers and compared them with the frozen pipeline output. For Manavgat this output is the one re-frozen on the corrected label (Section 2.2), produced with the pipeline at commit 6381f4c. Only two changes were made. A one-line patch lets the window-closure module accept a population column that the corrected label adds; its diff is released with the re-freeze. A runtime setting points the Earth Engine calls at the project `thermaltwin`, which changes only the infrastructure. The within-region results agree exactly, and the transfer directions to within 1.3×10^{-8}, below the repository tolerance of 10^{-6}. This tolerance applies to the CORAL and within-region check. The frame-transfer script of Section 3.4 fits its forests in parallel, so its values vary between runs by up to 2×10^{-6} (1.4×10^{-6} in a rerun on 29 September 2026, `paper/labelfix_rerun/round7/`). This is within the tolerance of 10^{-5} applied to it, and the values are stable at the printed precision. The effect of an unpinned library version is given in Section S3.5(vi).

S3.7 Transfer-gap decomposition and the concept-shift diagnostic, in full

For each direction, the gap between the within-region skill of the target and the raw transfer result is split in two parts. One part is what the best label-free adaptation recovers, and the other is what it does not recover. The recovered fraction is (adapted − raw) / (within − raw), signed and not clipped, with its interval from the same paired bootstrap. It describes what the two methods tested achieved, with the better one chosen using target labels; other correction methods were not tested. Section 3.3 shows that the remainder should not be read as a

concept-shift residual, because much of it already appears inside a single region (Section S1.10).

Concept shift is examined with the *signed univariate association*. For each numeric predictor, the raw ROC-AUC of the predictor against `burned` is computed in each region and is not folded to max(AUC, 1 − AUC), so a value below 0.5 shows the direction of the association. A reversal is called bootstrap-supported only when the point estimates of the two regions lie on opposite sides of 0.5 and the interval of each region excludes 0.5, under the same 10-cell spatial-block bootstrap. This is stricter than requiring disjoint intervals. A feature with disjoint intervals, one of which includes 0.5, has not been shown to point in any direction in that region, so it is recorded as a point reversal only. Section S2 repeats the rule next to the counts, and `conditional_similarity_transfer.json` stores it as metadata.

S3.7.1 Degenerate replicates

Replicates with $|G| < 10^{-6}$ are dropped.

S3.8 Label-free adaptation, full specification

Region-wise z-score. The numeric features of each region are standardised with the statistics of that region: source statistics from source data and target statistics from target data, never pooled. For feature j in region R,

$$z_{ij} = \frac{x_{ij} - \mu_j^R}{\sigma_j^R}, \qquad \mu_j^R = \frac{1}{n_j^R} \sum_{i \in O_j^R} x_{ij}\,, \qquad \sigma_j^R = \left(\frac{1}{n_j^R} \sum_{i \in O_j^R} (x_{ij} - \mu_j^R)^2\right)^{1/2}, \qquad (S4)$$

where O_j^R contains the n_j^R cells with an observed value (ddof 0). A missing value is first set to μ_j^R, so it becomes zero, and $\sigma_j^R < 10^{-12}$ is replaced by 1. Land cover is not transformed. Target feature statistics, but never target labels, are therefore used in the adapted models. The

classifier is refitted on the z-scored source and applied to the z-scored target. This removes differences in the mean and variance of each feature.

CORAL after region-wise z-score. The source covariance is aligned to the target covariance with the standard whitening and recolouring map (Sun et al. 2016). With Z_R the $n_R \times d$ matrix of z-scored numeric features, one row per cell,

$$Z_s^{\mathrm{al}} = Z_s\,(C_s + \lambda I)^{-1/2}\,(C_t + \lambda I)^{1/2}, \qquad C_R = \frac{1}{n_R}\sum_{i=1}^{n_R}(z_i - \bar{z}_R)(z_i - \bar{z}_R)^{\top}, \qquad (S5)$$

with $\lambda = 10^{-5}$ (ddof 0). Both means are zero after Eq. (S4), so the mean terms of the general map are zero. Matrix powers use a symmetric eigendecomposition, with eigenvalues set to at least 10^{-12}. The transform is applied to the source only. The target stays at Z_t, and the classifier is refitted on Z_s^{al}. Neither method uses a target label, and this is checked at run time. The sensitivity to λ was tested over nine values on four of the twenty directions; transfer AUC moved by at most 0.014, and λ was not selected on performance. λ = 1, the value of the original CORAL formulation, is tested in Sections S1.2 and S1.23. The value used throughout is $\lambda = 10^{-5}$.

S4 Target-label recovery curve

S4.1 Purpose

The main text shows that label-free alignment does not close the transfer gap. This section asks how much of the gap a small number of target labels can close. It is not a proposed method or an operational claim, because the labels come from the fire that is predicted. It covers three of the five regions (Manavgat, Bejís and Muğla) and all six directions among them. Source: the pipeline's few-shot recovery diagnostic on the corrected label, analysis

`7348dfe7…`, released in `paper/labelfix_rerun/exports/few_shot_recovery_7348dfe7/` (`recovery_curve.csv`, `repeat_metrics.csv`, `report.md`); every value below is read from it.

S4.2 Design

The unit of labelling effort is one 10-cell (≈5 km) spatial block of the target. For a budget of k blocks, k target blocks are drawn from the training folds of a 5-fold spatially blocked split of the target. Their labels are added to the source training set, and the refitted model is scored on the held-out target folds. The budgets are 0, 1, 2, 4, 8, 16 and 32 blocks, with ten repeated draws for every k > 0. Budget 0 is raw transfer. The *ceiling* is a target-only model at the same 10-cell blocking (0.777 for Muğla, 0.824 for Bejís and 0.882 for Manavgat). It reproduces the pipeline's own large-block values exactly. The *recovered fraction* is (few-shot − raw) / (ceiling − raw), the share of the gap between raw transfer and the ceiling that the labels close; it is signed and not clipped. The intervals are *selection intervals*: the 2.5th and 97.5th percentiles over the ten block draws. They describe only which blocks were drawn, and no hypothesis test is made.

S4.3 Result

Table S22. Few-shot recovery of target ROC-AUC, thermal model, natural-vegetation population. Raw = source-only transfer (budget 0); ceiling = target-only model at the same 10-cell blocking. Values are means over ten repeated block draws.

Direction	Raw	1 blk	2 blk	4 blk	8 blk	16 blk	32 blk	Ceiling	Gap closed at 32
Bejís → Manavgat	0.314	0.518	0.565	0.659	0.698	0.750	0.820	0.882	89 %
Muğla → Bejís	0.583	0.578	0.603	0.625	0.666	0.743	0.789	0.824	85 %
Manavgat → Bejís	0.396	0.447	0.456	0.541	0.598	0.690	0.752	0.824	83 %
Muğla → Manavgat	0.345	0.358	0.380	0.407	0.437	0.505	0.624	0.882	52 %
Manavgat → Muğla	0.438	0.447	0.453	0.468	0.490	0.532	0.571	0.777	39 %

Direction	Raw	1 blk	2 blk	4 blk	8 blk	16 blk	32 blk	Ceiling	Gap closed at 32
Bejís → Muğla	0.618	0.576	0.577	0.578	0.598	0.637	0.666	0.777	30 %

At 32 blocks, three of six directions close 83 to 89 % of the gap to the ceiling. Two of these three started below chance. The other three close 30 to 52 %. Thirty-two blocks contain 2,740 to 2,970 labelled cells, 7 to 20 % of the natural-vegetation population of the target. This is not a small budget. For Bejís, the 32 labelled blocks contain on average 880 of its 1,100 burned cells, and for Manavgat about 1,800 of 2,935.

Recovery is slow where transfer is worst. Muğla → Manavgat has the largest gap between raw transfer and the ceiling. It is still below 0.5 after 8 labelled blocks and closes 52 % at 32.

Small budgets harm the direction that already transfers. Bejís → Muğla is above chance without labels (0.618), and target labels help it least. The curve is below the raw value at 1, 2, 4 and 8 blocks (0.576 to 0.598), rises above it only at 16 blocks, and closes 30 % at 32.

S4.4 Limits

1. Three regions, six directions. Evia and Montiferru are not included, so this covers six of the twenty directions and cannot cover all five regions. Six directions cannot support a general label budget, so none is given.
2. It needs labelled target cells from the fire that is predicted. Nothing here is a label-free method, and nothing here changes the negative result for label-free alignment. Labels from earlier fires in the same region were not tested.
3. The interval is a selection interval. At the largest budgets it becomes narrow because the pool of blocks is used up, so its width says little about how well the value is estimated. Per outer fold, the target has on average about 12 blocks with both classes in Bejís, 29 in Manavgat and 48 in Muğla. At 16 and 32 blocks, the draws therefore

use almost all of these blocks and select almost the same set (`direction_budget_feasibility.csv`).

4. The ceiling is the 10-cell target-only value, which is lower than the ≈1 km within-region value of Table 2. The recovered fractions can only be read against this matched ceiling, and on the original study areas, which are not comparable across regions (Section 3.4).

S5 Code

The analysis code is public at https://github.com/dryuemco/thermal-twin under the MIT licence, and the version used for this paper is archived at Zenodo (https://doi.org/10.5281/zenodo.23070876). This section is a guide to it; the README of the repository gives the same information. The satellite processing pipeline is a separate public repository, https://github.com/emrehann17/satellite-thermal-digital-twin (MIT licence). The re-frozen Manavgat outputs were produced at its commit `6381f4c`. The outputs of the other regions were produced at earlier commits and are reproduced by it (Section S3.6.2).

Path	Role
`paper/code/`	Analysis and verification scripts. `_canonical.py` loads every modelling dataset, verifies its SHA-256 and asserts the leakage exclusions of Section 2.13 at every fit.
`paper/code/appendix_tables.py`	Rebuilds Tables S2 to S7 and S9 to S18 row by row from source files whose SHA-256 values are pinned in `paper/labelfix_rerun/round5/tables/SOURCES.sha256`, and compares them with this document.
`paper/figures/`	One script per figure. Each asserts every plotted value against the frozen outputs and the manuscript text, and checks its layout.
`paper/figures/check_all.py`	Runs all figure scripts and `appendix_tables.py`, restores the committed outputs byte for byte, and exits 0 when everything passes.
`paper/tex/`	Builds the manuscript and this document from their Markdown sources and checks the port: every number survives, and every cross-reference, citation and supplementary reference resolves.
`paper/labelfix_rerun/`	The corrected-label outputs from which every corrected number is read: `code/`, `geometry/`, `inference/`, `labels/` and `round3/` to `round8/` hold the analysis outputs, `pipeline/` the re-frozen pipeline outputs for Manavgat, and `exports/` the released pipeline diagnostics (few-shot recovery, CORAL λ sweep, window closure), each with its SHA-256 in `exports/SHA256SUMS.txt`.

All commands are run from the repository root. The environment is Python 3.12.10 with NumPy 2.4.4, pandas 3.0.2 and scikit-learn 1.9.0, fixed in `ENVIRONMENT.md`. Point estimates depend on the scikit-learn version (Section S3.5(vi)). Fig. 1 also needs cartopy; without it, `check_all.py` reports this figure as skipped. Tables S1, S8, S19 to S36 and the values in the text are not checked by `appendix_tables.py`; they name their source files in their captions or text.

S6 Data

All satellite inputs are public and were retrieved through Google Earth Engine (Table S20). No proprietary or restricted data were used.

The five modelling datasets are in the analysis repository, one per region, as `paper/data/<region>/step8a_500m_modeling_dataset.parquet`. Each is checked on load against the SHA-256 recorded in `paper/code/_canonical.py`. For Bejís, Muğla, North Evia and Montiferru this hash equals the pipeline's own record. For Manavgat the file is the corrected-label re-freeze (SHA-256 `5a5e876c…`; Section S3.1.2). The pipeline's record still names the original-label table (SHA-256 `054a1961…`), which is not in the repository and can be regenerated by the pipeline. The frozen numeric outputs behind every table and figure are in the same repository, identified by the paths and hashes given in the table captions and in `SOURCES.sha256`. The data derived from Landsat, MODIS, the Copernicus DEM and ESA WorldCover remain subject to the terms of these products, which are given in the data availability statement of the main article.

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